Distributionally Robust Fault Detection for Stochastic Linear Time-Varying Systems with Unknown Inputs (Extended Version)

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Abstract

A key limitation of conventional statistical fault detection (FD) is its assumption of perfectly known noise distributions, which compromises robustness against inexact distribution information. While recent progress has addressed such distributional ambiguity, it remains limited to linear time-invariant systems. This paper proposes a distributionally robust FD method for stochastic linear time-varying systems subject to both distributional ambiguity and unknown inputs. Using the moment-based distributional ambiguity set, the proposed method proceeds in two stages. First, the residual signal is generated to decouple unknown inputs while ensuring temporal uncorrelation, even under distributional ambiguity. Second, a quadratic FD test is designed via distributionally robust optimization. The worst-case fault detection rate (FDR) is maximized while constraining the worst-case false alarm rate (FAR). The resulting FD test retains the same form as the classical generalized likelihood ratio (GLR) test, with only its threshold adjusted to compensate for distributional ambiguity. Moreover, a closed-form expression of worst-case FDR is derived for any given fault, motivating the use of a low-order fault approximation to further improve FDR. Simulations on a synthetic example and a quadruple water tank system demonstrate that the proposed approach achieves a superior FAR-FDR trade-off compared to the GLR test.

Key words: Fault detection, stochastic linear time-varying systems, unknown input, distributional ambiguity, distributionally robust optimization.

1 Introduction

The timely and accurate detection of anomalies or faults is of paramount importance for modern safety-critical systems. Over recent decades, model-based fault detection (FD) has received considerable attention and has been applied across aerospace, automotive, and chemical industries, see e.g. [3,7,29,30,48] and reference therein. A primary objective in FD is to achieve a high fault detection rate (FDR) while maintaining a low false alarm rate (FAR), in the presence of various uncertainties such as external disturbances or model inaccuracies [1,7]. However, raising the FDR typically comes at the expense of a higher FAR. Consequently, designing an effective FD system necessitates an optimal trade-off between FDR and FAR [1,7].

Based on how uncertainties are characterized, existing robust FD methods can be broadly classified into two categories. The first category, deterministic FD methods, employ a worst-case or set-theoretic approach to cope with unknown-but-bounded disturbances and model uncertainties [7,24,33,44]. These approaches maximize fault sensitivity (expressed in various forms such as system norms or gap metrics) while ensuring zero FAR via an appropriately designed detection threshold. As false alarms must be completely avoided under all scenarios including worst-case uncertainties with vanishingly low probability, the resulting FDR of such methods may be severely compromised.

In contrast to the deterministic approaches, the statistical FD methods assume various uncertainties to be stochastic. By leveraging the underlying probability dis-

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tributions of uncertainties that distinguish between low- and high-probability events, these methods can establish a far less conservative detection threshold. This is accomplished by allowing an acceptable FAR, which in turn leads to a significantly improved FDR. For linear dynamic systems with additive Gaussian noises, the typical statistic FD method firstly employs a Kalman filter or a parity relation to generate a whitened residual signal. It then analyzes how the considered fault alters the statistical properties of this residual (e.g., its mean or covariance), and finally detects that change using generalized likelihood ratio (GLR) test [9,23,31,38,45]. For stochastic nonlinear systems, the GLR test have been integrated with techniques such as extended Kalman filtering [19], moving horizon estimation [18], and particle filtering [4]. A more recent development is stochastic active fault diagnosis, which aims at designing the input signal to actively reduce the misdiagnosis probability [21,22]. Meanwhile, several extended deterministic FD approaches have been put forward to incorporate information from probability distributions of parametric uncertainties or disturbances. In the observer-based FD scheme, the randomized algorithm is leveraged by a probabilistic relaxation of the norm-based threshold computation [50] and the FAR-constrained optimal design [8,10]. Similarly, the set-membership approach is generalized in [5,42] by admitting a risk level in the set-based consistency test.

When the monitored system is subject not only to stochastic noises but also to non-random uncertainties or nuisance faults whose probability distributions are entirely unknown, the aforementioned statistical FD methods may fail to ensure an acceptable FAR. This limitation has motivated the research on robust statistical FD method to mitigate the adversary effects of such nuisance uncertainties/faults. For stochastic linear systems, one line of work introduces additive unknown inputs to model the influence of nuisance uncertainties/faults, and then modifies the residual generator such that the GLR test can be applied to the residual signal decoupled from the unknown inputs [13,14]. Alternatively, a constrained GLR test approach is proposed in [25] to leverage the bounded information of these unknown inputs. In parallel, deterministic FD methods have extensively investigated unknown input rejection in the FD observer design [40,47], though typically ignoring probability distributions of uncertainties/noises.

Most statistical hypothesis testing and FD methods rely on the assumption that the probability distributions of stochastic uncertainties are exactly known. However, it is common in practice that the true distributions deviate from the assumed or estimated ones, which can severely degrade detection performance. To address this issue in the case of *temporally independent* observations, there has been growing research interest on robust hypothesis test that minimizes the worst-case weighted sum of error probabilities over sets of partially known distributions described by divergence [11,20] or Wasserstein distance metrics [15]. Due to the requirement of temporally independent observations, these robust hypothesis test methods cannot be directly applied to the raw observations of a dynamic system. For dynamical systems, while robustness against deterministic uncertainties or unknown inputs has been extensively studied in FD literature, the problem of robustness against inexact distribution information has received considerably less attention so far in statistical FD. To address the issue of imprecise distribution knowledge, some recent FD literature adopted the concept of distributional ambiguity sets which are constructed using moments [34,37,39,49], Kullback-Leibler divergence [41], or Wasserstein metrics [12,36]. Subsequently, distributionally robust chance constrained optimization problems are formulated and solved for the FD test design.

It is worth noting that all aforementioned distributionally robust FD literature are confined to stochastic linear time-invariant (LTI) systems. This limitation motivates us to investigate the distributionally robust FD problem for stochastic linear time-varying (LTV) systems in this paper, as the LTV model provides a simple yet powerful approximation to practical nonlinear systems [48]. Specifically, we propose a distributionally robust FD method for stochastic LTV systems that are subject to both distributional ambiguity and non-random unknown inputs. Using the moment-based ambiguity set consisting of all noise distributions with the specified mean and covariance, the proposed method proceeds in two main steps. First, a residual generator rejecting unknown inputs is put forward. It generates a temporally uncorrelated residual signal whose statistical independence at different times cannot be guaranteed due to distributional ambiguity. Second, a distributionally robust optimization problem is formulated to construct the FD test in a quadratic form. The objective is to maximize the worst-case FDR, while imposing a distributionally robust chance constraint to maintain the worst-case FAR below a desired probability level. Compared to the aforementioned distributionally robust FD literature, our work makes the following contributions:

i) Our proposed distributionally robust FD design considers stochastic LTV systems affected by both distributional ambiguity and non-random unknown inputs. In contrast, existing distributionally robust FD literature are confined to LTI systems without unknown inputs.
ii) Our formulated distributionally robust design problem directly maximizes the worst-case FDR under the worst-case FAR constraint. The resulting solution is derived in closed form, and elegantly retains the same structure as the classical GLR test, with only the detection threshold adjusted to compensate for distributional ambiguity. The obtained closed-form solution can be efficiently computed by a parallel recursive algorithm. In con-

trast, existing distributionally robust FD methods maximize a fault sensitivity metric as an indirect means to enhance FDR [36,37].

iii) We have analytically derived a closed-form expression of the worst-case FDR under a given fault signal. Built on this result, our analysis reveals that employing a low-order fault approximation can further improve the FDR. This strategy is unexplored in existing distributionally robust FD literature.

This paper is structured as follows. For the problem setting described in Section 2, residual generation is investigated in Section 3, and the distributionally robust FD problem is formulated in Section 4. The resulting solution is derived in Section 5. Section 6 reports case studies using a numerical example and a quadruple water tank system. Concluding remarks are provided in Section 7.

Notations: The space of symmetric matrices of dimension n is denoted by $\mathbb{S}^n$. For a square matrix X, $\mathrm{tr}(X)$ represents the trace of X. For $X \in \mathbb{S}^n$, $X \succ 0$ ($X \succeq 0$) implies that X is positive definite (positive semidefinite). The square root of a positive semidefinite matrix X is denoted by $X^{\frac{1}{2}}$. The column space of a matrix X is denoted by $\mathrm{col}(X)$. For matrices $X_1, X_2, \cdots, X_n$, $\mathrm{diag}(X_1, X_2, \cdots, X_n)$ represents a block-diagonal matrix whose diagonal blocks are $X_1, X_2, \cdots, X_n$. An n-dimensional vector of all ones is denoted by $\mathbb{1}_n$. An n-dimensional identity matrix is denoted by I_n. For a vector $x \in \mathbb{R}^n$ and a symmetric matrix $A \in \mathbb{S}^n$, let $\|x\| = \sqrt{x^\top x}$ and $\|x\|_A^2 = \sqrt{x^\top A x}$ represent the 2-norm and the weighted 2-norm of x, respectively. A zero vector or matrix is denoted by $\mathbf{0}$ whose size is determined in context if not specified.

2 Problem statement

Consider the problem of detecting an additive fault in a stochastic linear time-varying (LTV) system

$$\begin{aligned} x_{k+1} &= A_k x_k + B_k u_k + E_k d_k + \Psi_k(\rho)\theta + w_k \\ y_k &= C_k x_k + v_k, \end{aligned} \tag{1}$$

where $x_k \in \mathbb{R}^{n_x}$ is the system state, $u_k \in \mathbb{R}^{n_u}$ is the control input, $y_k \in \mathbb{R}^{n_y}$ is the measured output, $d_k \in \mathbb{R}^{n_d}$ is the unknown input signal to be rejected in FD, $\theta \in \mathbb{R}^{n_\theta}$ represents a constant fault vector, w_k and v_k denote mutually independent white noises, and all system matrices in (1) are known, time-varying and bounded. As in [45], the fault term $\Psi_k(\rho)\theta$ in (1) consists of a time-varying fault profile matrix $\Psi_k(\rho)$ and a fault vector θ, with ρ being the unknown fault onset time. Specifically, we consider the step fault in this paper, i.e.,

$$\Psi_k(\rho) = \begin{cases} 0 & k < \rho, \\ \widetilde{\Psi}_k & k \geq \rho. \end{cases} \tag{2}$$

The statistical FD methods typically require the distribution information of the white noises w_k and v_k. In particular, existing statistical FD methods, e.g, the Kalman filter (KF) based GLR test, for stochastic LTV systems assume the noises to be white and Gaussian [45]. In practice, however, the noise distributions $\mathbb{P}_{w_k}$ and $\mathbb{P}_{v_k}$ are usually unknown a priori. Reliable probability density estimation often requires a large sample size, especially for a multivariate random variable [35]. Moreover, it is inevitable that the actual noise distribution deviates from the estimated one. This could lead to a high FAR or a low FDR if it is not properly taken into account.

To cope with such uncertainty in the distribution information, a moment-based ambiguity set $\mathcal{P}(\mu, \Sigma)$ is introduced in [36,41,49] to describe all probability distributions with given mean μ and covariance $\Sigma \succeq 0$. This requires only accurate estimation of mean and covariance from data of a moderate sample size, which is significantly easier than probability density estimation.

Assumption 1 *The initial state x_0 and white noises w_k and v_k are mutually independent, and their distributions belong to the moment-based ambiguity sets $\mathcal{P}(\bar{x}_0, P_0^x)$ $\mathcal{P}(\mathbf{0}, Q_k)$ and $\mathcal{P}(\mathbf{0}, R_k)$, respectively.*

Assumption 2 *The conditions $C_k E_{k-1} \neq \mathbf{0}$ and $\mathit{rank}(C_k E_{k-1}) = \mathit{rank}(E_{k-1}) < n_y$ hold for all $k > 0$.*

As in [16,17,26], Assumption 2 is the necessary and sufficient condition to allow perfect unknown input decoupling for residual generation later in Section 3.

Considering the stochastic LTV system (1) with both distributional ambiguity and unknown inputs in Assumptions 1 and 2, we focus on a distributionally robust design of the FD test which both rejects unknown inputs and maximizes the worst-case FDR while maintaining the worst-case FAR below a desired probability level.

3 Residual generation rejecting unknown inputs

The classical statistical FD method generally performs the standard GLR or cumulative sum test on a whitened residual signal generated from a KF. This, however, cannot be applied to our considered problem described in Section 2 due to the presence of unknown input and distributional ambiguity. In particular, it is impossible for the generated residual to be strictly white under distributional ambiguity. In this section, we will first construct an unknown input decoupled filter that generates an uncorrelated residual signal (not necessarily statistically independent at different times), and then analyze the fault effect in the residual. Such a residual generation filter serves as the basis for our proposed distributionally robust FD test in the next section.

3.1 Residual generation filter design

Inspired by [17,43], we construct the unknown input decoupled residual generation filter in the form

$$\begin{aligned}\hat{x}_{k|k} &= A_{k-1}\hat{x}_{k-1|k-1} + B_{k-1}u_{k-1} \\ &\quad + L_k(y_k - C_kA_{k-1}\hat{x}_{k-1|k-1} - C_kB_{k-1}u_{k-1}), \\ r_k &= \Pi_k(y_k - C_kA_{k-1}\hat{x}_{k-1|k-1} - C_kB_{k-1}u_{k-1}),\end{aligned} \tag{3}$$

where $\hat{x}_{i|j}$ represents the state estimate of x_i given measurements up to time j, and the residual signal r_k is computed from the one-step output prediction. The filter gain L_k and the post-processing matrix Π_k are to be designed. According to (1) and (3), the dynamic evolution of the state estimation error $\tilde{x}_{k|k} \triangleq x_k - \hat{x}_{k|k}$ is governed by

$$\begin{aligned}\tilde{x}_{k|k} &= (I_{n_x} - L_kC_k)(A_{k-1}\tilde{x}_{k-1|k-1} + w_{k-1}) - L_kv_k \\ &\quad + (I_{n_x} - L_kC_k)\left[E_{k-1}d_{k-1} + \Psi_{k-1}(\rho)\theta\right] \\ r_k &= \Pi_k(C_kA_{k-1}\tilde{x}_{k-1|k-1} + C_kw_{k-1} + v_k) \\ &\quad + \Pi_kC_k\left[E_{k-1}d_{k-1} + \Psi_{k-1}(\rho)\theta\right].\end{aligned} \tag{4}$$

For the subsequent FD, the selection of L_k and Π_k in (3) aims at achieving the following objectives:

i) Unknown input decoupling in the residual dynamics, i.e.,

$$L_kC_kE_{k-1} = E_{k-1}, \tag{5a}$$
$$\Pi_kC_kE_{k-1} = 0, \tag{5b}$$

according to the residual dynamics (4);

ii) Unbiased minimum-variance (UMV) estimation in the fault-free case, i.e.,

$$\begin{aligned}&\min_{L_k} \mathbb{E}_{\tilde{x}_{k-1|k-1}, w_{k-1}, v_k} \left\|\tilde{x}_{k|k}\right\|_2^2 \\ &\text{s.t. } L_kC_kE_{k-1} = E_{k-1}\end{aligned} \tag{6}$$

with its constraint from (5a);

iii) Temporal uncorrelation of the fault-free component $r_{s,k}$ of the generated residual signal r_k, i.e.,

$$\mathbb{E}\{r_{s,k}r_{s,k+i}^\top\} = 0, \quad \forall k > 0, i > 0. \tag{7}$$

While (5) and (6) are standard for unknown input decoupled state estimation and residual generation as in [16,17], the temporal uncorrelation requirement (7) is dedicated to our proposed distributionally robust FD design later in Section 4. Note that unlike the conventional statistical FD test in the Gaussian case, it is impossible to whiten the fault-free residual component due to the distributional ambiguity in Assumption 1.

To satisfy the exact decoupling condition (5), there exist $L_k \neq 0$ and $\Pi_k \neq 0$ if and only if Assumption 2 holds. This does not require C_kE_{k-1} or E_{k-1} to have full column rank, because we just need to decouple the effect of the unknown input d_k rather than estimating d_k. To parametrize L_k satisfying (5a), we introduce the full rank decomposition of E_{k-1} in the form

$$E_{k-1} = G_{k-1}H_{k-1}, \tag{8}$$

with $G_{k-1} \in \mathbb{R}^{n_x \times n_e}$, $H_{k-1} \in \mathbb{R}^{n_e \times n_d}$, and n_e being the rank of E_{k-1}. Then it is straightforward to see that both C_kG_{k-1} and G_{k-1} have full column rank according to Assumption 2. Since H_{k-1} is of full row rank, the decoupling condition (5a) is equivalent to $L_kC_kG_{k-1} = G_{k-1}$, hence L_k satisfying (5a) can be parameterized as

$$L_k = G_{k-1}M_k + K_k(I_{n_y} - C_kG_{k-1}M_k), \tag{9}$$

where K_k is to be determined later, and M_k is the left inverse of C_kG_{k-1}, i.e.,

$$M_kC_kG_{k-1} = I_{n_e}. \tag{10}$$

Similarly, Π_k needs to satisfy $\Pi_kC_kG_{k-1} = 0$ according to (5b) and (8), thus it is parameterized as

$$\Pi_k = \Xi_k(I_{n_y} - C_kG_{k-1}M_k), \tag{11}$$

with Ξ_k to be determined later.

Next, we consider the UMV estimation problem (6). Since deriving the solution to this problem requires only mean and covariance information, its optimal solution with respect to the moment-based distributional ambiguity described in Assumption 1 (with exact mean and covariance) is exactly the same as that given by [17] in the conventional Gaussian setting. Below we briefly summarize the optimal solution, and the derivation details are referred to [17].

Using (9)–(11), the error dynamics (4) in the fault-free case ($\theta = 0$) can be rewritten as

$$\begin{aligned}\tilde{x}_{s,k|k} &= (I_{n_x} - K_kC_k)(A_{k-1}^*\tilde{x}_{s,k-1|k-1} + w_{k-1}^*) - K_kv_k \\ &= (I_{n_x} - K_kC_k)z_{s,k} - K_kv_k, && \text{(12a)} \\ r_{s,k} &= \Xi_k(C_kz_{s,k} + v_k), && \text{(12b)} \\ z_{s,k} &= A_{k-1}^*\tilde{x}_{s,k-1|k-1} + w_{k-1}^*, && \text{(12c)} \\ A_{k-1}^* &= (I_{n_x} - G_{k-1}M_kC_k)A_{k-1}, && \text{(12d)} \\ w_{k-1}^* &= (I_{n_x} - G_{k-1}M_kC_k)w_{k-1} - G_{k-1}M_kv_k, && \text{(12e)}\end{aligned}$$

where $\tilde{x}_{s,k|k}$ and $r_{s,k}$ represent the fault-free components of the state estimation error $\tilde{x}_{k|k}$ and the residual r_k, respectively. Despite the distributional ambiguity described in Assumption 1, the initial state covariance P_0^x as well as the noise covariances $\{Q_k, R_k\}$ are accurate, thus the dynamic evolution of error covariance $P_{k|k} \triangleq \mathbb{E}\{\tilde{x}_{s,k|k}\tilde{x}_{s,k|k}^\top\}$ can be exactly expressed as

$$P_{k|k} = K_kR_k^*K_k^\top - V_k^*K_k^\top - K_k(V_k^*)^\top + P_k^*, \tag{13}$$

with

$$
\begin{aligned}
Q_{k-1}^* &= \mathbb{E}_{w_{k-1},v_k}\left[w_{k-1}^* w_{k-1}^{*\top}\right] \\
&= (I_{n_x} - G_{k-1}M_kC_k)Q_{k-1}(I_{n_x} - G_{k-1}M_kC_k)^\top \\
&\quad + G_{k-1}M_kR_kM_k^\top G_{k-1}^\top, \qquad (14a)\\
P_k^* &= \mathbb{E}[z_{s,k}z_{s,k}^\top] = A_{k-1}^* P_{k-1|k-1}A_{k-1}^{*\top} + Q_{k-1}^*, \qquad (14b)\\
S_k^* &= \mathbb{E}\left[z_{s,k}v_k^\top\right] = \mathbb{E}\left[w_{k-1}^* v_k^\top\right] = -G_{k-1}M_kR_k, \qquad (14c)\\
R_k^* &= \mathbb{E}\left[(C_kz_{s,k}+v_k)(C_kz_{s,k}+v_k)^\top\right] \\
&= C_kP_k^*C_k^\top + C_kS_k^* + (C_kS_k^*)^\top + R_k \qquad (14d)\\
&= (I_{n_y} - C_kG_{k-1}M_k)\widetilde{R}_k(I_{n_y} - C_kG_{k-1}M_k)^\top, \\
\tilde{R}_k &= C_k(A_{k-1}P_{k-1|k-1}A_{k-1}^\top + Q_k)C_k^\top + R_k, \qquad (14e)\\
V_k^* &= \mathbb{E}[z_{s,k}(C_kz_{s,k}+v_k)^\top] = P_k^*C_k^\top + S_k^*. \qquad (14f)
\end{aligned}
$$

Using (9) to replace L_k, the UMV estimation problem (6) can be equivalently expressed as $\min_{K_k}\ \mathrm{trace}(P_{k|k})$, i.e.,

$$
\min_{K_k}\ \mathrm{trace}(K_kR_k^*K_k^\top - V_k^*K_k^\top - K_kV_k^{*\top} + P_k^*) \qquad (15)
$$

according to (13), and the equality constraint in (6) is automatically satisfied. As indicated in Lemma 6 of [17], $I_{n_y} - C_kG_{k-1}M_k$ does not have full row rank, hence R_k^* is rank-deficient according to (14d) and (14e). Consequently, the solution to the above UMV problem (15) is non-unique, which can be parametrized as follows.

Lemma 1 (Theorem 7 in [17]) *Let $m < n_y$ represent the rank of R_k^*. There exists a matrix $\alpha_k \in \mathbb{R}^{m\times n_y}$ such that*

$$
\mathit{rank}(\alpha_kR_k^*\alpha_k^\top) = \mathit{rank}(R_k^*) = m \qquad (16)
$$

holds. Then the optimal gain matrix K_k solving the UMV problem (15) is

$$
K_k = V_k^*\alpha_k^\top(\alpha_kR_k^*\alpha_k^\top)^{-1}\alpha_k. \qquad (17)
$$

Finally, we proceed to the temporal uncorrelation requirement (7) by selecting the post-process matrix Π_k in (4). This is equivalent to determining Ξ_k in (11). Since the noises w_k and v_k have the moment-based distributional ambiguity as described in Assumption 1, the obtained fault-free residual component $\{r_{s,k}\}$ may not be statistically independent at different times although it can be made statistically uncorrelated as follows.

Theorem 1 *Consider the moment-based distributional ambiguity in Assumption 1 and the fault-free case, i.e., $\theta = 0$. Having L_k in (9), K_k in (17), and*

$$
\Xi_k = \alpha_k, \qquad (18)
$$

the unknown input decoupled filter (3) generates a residual signal whose fault-free component $r_{s,k}$ is zero mean and statistically uncorrelated. In particular, the distribution of $r_{s,k}$ belongs to the ambiguity set $\mathcal{P}(0, \alpha_kR_k^*\alpha_k^\top)$, and

$$
\mathbb{E}[r_{s,k}r_{s,k+i}^\top] = \alpha_kR_k^*\alpha_k^\top\delta_{k,k+i}, \qquad (19)
$$

with $\delta_{m,n} = 1$ if $m = n$ and $\delta_{m,n} = 0$ if $m \neq n$.

The proof is given in Appendix A.

The residual generation filter (3) derived above is recursively computed as in Lines 4 $\sim$ 10 of Algorithm 1. In such recursions, M_k and α_k are chosen to satisfy (10) and (16), which are non-unique.

3.2 Fault effect in the residual

Next, we follow the idea of Section 3.1 in [45] to analyze fault effect in the residual r_k generated from the UMV filter derived in Section 3.1, with an extension to cope with unknown inputs. Both $\tilde{x}_{k|k}$ and r_k in the error dynamics (4) are expressed as

$$
\begin{aligned}
\tilde{x}_{k|k} &= \tilde{x}_{s,k|k} + \Gamma_k(\rho)\theta, \qquad (20a)\\
r_k &= r_{s,k} + \Phi_k(\rho)\theta, \qquad (20b)
\end{aligned}
$$

where $\tilde{x}_{s,k|k}$ and $r_{s,k}$ are the fault-free components defined in (12), and $\Gamma_k(\rho)$ and $\Phi_k(\rho)$ are the fault profile matrices given in the following proposition, with ρ being the unknown fault onset time.

Proposition 1 *With L_k in (9), Π_k in (11), K_k in (17), and $\Xi_k = \alpha_k$, the fault profile matrices $\Gamma_k(\rho)$ and $\Phi_k(\rho)$ in (20) are recursively updated as follows:*

$$
\begin{aligned}
\Gamma_k(\rho) &= (I_{n_x} - K_kC_k)(A_{k-1}^*\Gamma_{k-1}(\rho) + \Psi_{k-1}^*(\rho)), \qquad (21a)\\
\Phi_k(\rho) &= \alpha_kC_k(A_{k-1}^*\Gamma_{k-1}(\rho) + \Psi_{k-1}^*(\rho)), \qquad (21b)
\end{aligned}
$$

with A_{k-1}^* defined in (12d) and

$$
\begin{aligned}
&\Psi_{k-1}^*(\rho) = (I_{n_x} - G_{k-1}M_kC_k)\Psi_{k-1}(\rho), \\
&\Psi_k(\rho) = 0,\ \Gamma_k(\rho) = 0,\ \ \Phi_k(\rho) = 0,\ \textit{for } k < \rho. \qquad (22)
\end{aligned}
$$

The proof is given in Appendix B.

In summary, the obtained residual generator (3) has the same gain matrix L_k as the unknown input decoupled state estimator given in [6,17], and relies on the post-processing matrix Π_k to satisfy the decoupling condition (5b) and the temporal uncorrelation condition (7). As such, by following [46], we can prove that the residual generator (3) has exponential stability as well as

boundedness of the error covariance $P_{k|k}$ and fault profile matrix $\Gamma_k(\rho)$ if (A_k^*, C_k) is uniformly completely observable and $(A_k, Q_k^{\frac{1}{2}})$ is uniformly completely controllable. The basic idea of the proof is briefly described as follows: the error dynamics (12a) is equivalent to that of the conventional KF designed for the linear system $x_{k+1}^a = A_k^* x_k^a + w_k^*, y_k^a = C_k x_k^a + v_k$ without unknown inputs, hence existing results on stability and boundedness of the LTV KF can be used to obtain the sufficient condition for stability and boundedness of our proposed residual generator (3). The detailed proof is similar to the analysis in [46], which is omitted due to limited space.

4 Distributionally robust FD problem

With the aid of the residual generator derived in Section 3.1, the fault effect is additive in the residual signal, as indicated in (20b). Hence the FD problem is formulated as a statistical test between the following two hypotheses:

$$\begin{aligned}\mathbb{H}_0 &: r_k \sim \mathbb{P}_{r_k}^0 \in \mathcal{P}(0, \alpha_k R_k^* \alpha_k^\top),\\ \mathbb{H}_1 &: r_k \sim \mathbb{P}_{r_k}^1 \in \mathcal{P}(\Phi_k(\rho)\theta, \alpha_k R_k^* \alpha_k^\top),\end{aligned}$$

where $\mathbb{P}_{r_k}^i$ represents the residual distribution under the hypothesis $\mathbb{H}_i$ ($i = 0$ or 1), and $\mathcal{P}(\mu, \Sigma)$ represents a distributional ambiguity set consisting of distributions with the specified mean μ and covariance Σ.

To account for the unknown fault onset time ρ, we consider each possible onset time ρ within a sliding window $[k-l+1, k)$ of a predefined length l as in [45], and construct the following FD test:

$$\max_{k-l+1\le\rho<k} J(\mathbf{r}_\rho^k) \begin{cases} \le 1 \Rightarrow \text{ no fault alarm},\\ > 1 \Rightarrow \text{ fault alarm},\end{cases} \tag{23}$$

where $\mathbf{r}_\rho^k$ is the stacked residual vector defined as

$$\mathbf{r}_\rho^k = \begin{bmatrix} r_{\rho+1}^\top & r_{\rho+2}^\top & \cdots & r_k^\top \end{bmatrix}^\top, \tag{24}$$

and $J(\mathbf{r}_\rho^k)$ is the residual evaluation function in the quadratic form

$$J(\mathbf{r}_\rho^k) = \left\|\mathbf{r}_\rho^k\right\|_{\mathbf{W}_\rho^k}^2, \tag{25}$$

with the symmetric weighting matrix $\mathbf{W}_\rho^k \succeq 0$ to be designed. Note that the FD test (23) uses a normalized threshold.

The above problem setting, however, has two features preventing the use of standard GLR or cumulative sum tests: i) the distribution information under each hypothesis is inexact; and ii) the fault-free noise signal $r_{s,k}$ is not statistically independent at different times. To address this issue, we formulate the distributionally robust design problem as follows for the FD test (23) with ρ being fixed to a specific value within the range $[k-l+1, k)$.

Firstly, according to (20b), express the stacked residual $\mathbf{r}_\rho^k$ in (24) as

$$\mathbf{r}_\rho^k = \boldsymbol{\xi}_\rho^k + \mathbf{F}_\rho^k \theta, \tag{26}$$

where the fault-free component $\boldsymbol{\xi}_\rho^k$ and the fault profile matrix $\mathbf{F}_\rho^k$ are defined as

$$\begin{aligned}\boldsymbol{\xi}_\rho^k &= \begin{bmatrix} r_{s,\rho+1}^\top & r_{s,\rho+2}^\top & \cdots & r_{s,k}^\top \end{bmatrix}^\top,\\ \mathbf{F}_\rho^k &= \begin{bmatrix} \Phi_{\rho+1}^\top(\rho) & \Phi_{\rho+2}^\top(\rho) & \cdots & \Phi_k^\top(\rho) \end{bmatrix}^\top.\end{aligned} \tag{27}$$

Let $\mathbb{P}_{\xi_\rho^k}$ denote the probability distribution of $\boldsymbol{\xi}_\rho^k$, then it follows from the statistical uncorrelation of the fault-free residual component as stated in Theorem 1 that

$$\begin{aligned}&\mathbb{P}_{\xi_\rho^k} \in \mathcal{P}(0, \boldsymbol{\Sigma}_\rho^k) \text{ with}\\ &\boldsymbol{\Sigma}_\rho^k = \operatorname{diag}\{\alpha_{\rho+1} R_{\rho+1}^* \alpha_{\rho+1}^\top, \cdots, \alpha_k R_k^* \alpha_k^\top\}.\end{aligned} \tag{28}$$

Definition 1 *If the FD test* (23) *fixes the fault onset time ρ to be a specific value, the resulting FAR and FDR are defined as*

$$\text{FDR} = h(\mathbb{P}_{\xi_\rho^k}, \theta, \mathbf{W}_\rho^k), \ \textit{for } \theta \neq \mathbf{0}, \tag{29a}$$

$$\text{FAR} = h(\mathbb{P}_{\xi_\rho^k}, \mathbf{0}, \mathbf{W}_\rho^k), \tag{29b}$$

with

$$\begin{aligned}h(\mathbb{P}_{\xi_\rho^k}, \theta, \mathbf{W}_\rho^k) &= \mathbb{P}_{\xi_\rho^k}\left\{J(\mathbf{r}_\rho^k) > 1\right\}\\ &= \mathbb{P}_{\xi_\rho^k}\left\{\left\|\mathbf{F}_\rho^k\theta + \boldsymbol{\xi}_\rho^k\right\|_{\mathbf{W}_\rho^k}^2 > 1\right\}.\end{aligned} \tag{30}$$

Consider the distributional ambiguity set in (28), the worst-case FAR is $\sup_{\mathbb{P}_{\xi_\rho^k} \in \mathcal{P}(\mathbf{0}, \boldsymbol{\Sigma}_\rho^k)} h(\mathbb{P}_{\xi_\rho^k}, \mathbf{0}, \mathbf{W}_\rho^k)$ according to (30). Similarly, $\inf_{\theta\in\Theta, \mathbb{P}_{\xi_\rho^k} \in \mathcal{P}(\mathbf{0}, \boldsymbol{\Sigma}_\rho^k)} h(\mathbb{P}_{\xi_\rho^k}, \theta, \mathbf{W}_\rho^k)$ represents the worst-case FDR over a set of fault vectors Θ. For the FD test (25) with a given ρ, the distributionally robust design aims at maximizing the worst-case FDR while ensuring that the worst-case FAR is below a desired probability level γ. This design object is formulated as the maximin optimization problem with a chance constraint, i.e.,

$$\begin{aligned}\max_{\mathbf{W}_\rho^k \succeq 0} \ &\inf_{\theta\in\Theta, \mathbb{P}_{\xi_\rho^k} \in \mathcal{P}(\mathbf{0}, \boldsymbol{\Sigma}_\rho^k)} h(\mathbb{P}_{\xi_\rho^k}, \theta, \mathbf{W}_\rho^k)\\ \text{s.t.} \ &\sup_{\mathbb{P}_{\xi_\rho^k} \in \mathcal{P}(\mathbf{0}, \boldsymbol{\Sigma}_\rho^k)} h(\mathbb{P}_{\xi_\rho^k}, \mathbf{0}, \mathbf{W}_\rho^k) \le \gamma.\end{aligned} \tag{31}$$

where the fault set Θ is defined as

$$\Theta = \left\{ \theta \,\middle|\, \left\| \mathbf{F}_\rho^k \theta \right\|_{(\mathbf{\Sigma}_\rho^k)^{-1}} \geq \delta \right\}. \tag{32}$$

In (32), the 2-norm of $\mathbf{F}_\rho^k\theta$ is inversely weighted by the noise covariance matrix $\mathbf{\Sigma}_\rho^k$, and δ is predefined to be sufficiently large such that the worst-case FDR $\inf_{\theta\in\Theta, \mathbb{P}_{\xi_\rho^k}\in\mathcal{P}(\mathbf{0},\mathbf{\Sigma}_\rho^k)} h(\mathbb{P}_{\xi_\rho^k}, \theta, \mathbf{W}_\rho^k)$ in the cost function of (31) is nonzero. Note that in the case with Gaussian noises, $\left\| \mathbf{F}_\rho^k\theta \right\|_{(\mathbf{\Sigma}_\rho^k)^{-1}}$ is equal to the Kullback-Leibler divergence between the fault-free and faulty Gaussian residual distributions, i.e., $\mathcal{N}(0, \mathbf{\Sigma}_\rho^k)$ and $\mathcal{N}(\mathbf{F}_\rho^k\theta, \mathbf{\Sigma}_\rho^k)$. More discussions on the selection of δ will be given later in Section 5.

The above problem (31) is formulated for a hypothesized fault onset time ρ within the sliding window $[k-l+1, k]$. As will be seen at the end of Section 5.1, the optimal solution to (31) is expressed in a closed form, and an efficient recursive algorithm can be derived for its real-time computation. This attractive feature also enables extension to address the unknown fault onset time ρ by considering all possible values of ρ within $[k-l+1, k]$, as will be explained in Section 5.5.

Remark 1 *As will be shown in Section 5, the definition of the fault set Θ in (32) enables deriving a closed-form analytical solution that can be recursively computed. If the fault set Θ is in a different form, it would be extremely difficult, if not impossible, to have an analytic solution, thus would require more expensive online optimization which is left to future research.*

Remark 2 *The distributionally robust FD problem has recently attracted increasing attention in literature [12,34,36,37,39,41,49]. The ambiguity sets for inexact noise distributions are described by moments [34,37,39,49], Kullback-Leibler divergence [41], or Wasserstein metrics [12,36]. In [34], a distributionally robust threshold design is proposed for a given residual evaluation function to guarantee a desired level of worst-cased FAR. In [39,49], the distributionally robust parity space based designs select both the parity vector and the detection threshold to maximize a weighted sum of worst-case classification accuracy rates, which are limited to residual scalars. Retaining the advantage of multi-dimensional residual vectors, the proposed distributionally robust method in [12,36,37] choose the quadratic residual evaluation function to maximize a fault sensitivity metric under the worst-case FAR constraint. However, enhancing fault sensitivity does not necessarily imply a higher FDR (See Chapter 12 in [7]). In contrast to [36,37], our problem formulation (31) directly maximizes the worst-case FDR under the FAR constraint by choosing the quadratic residual evaluation function (25) with respect to the normalized threshold. Moreover, we consider the LTV system (1) with unknown inputs whilst the above distributionally robust FD literature are confined to LTI systems without non-random unknown inputs.*

5 Solution to distributionally robust FD

In this section, we first solve the chance constrained maximin optimization problem (31) with a given fault onset time ρ. Then, we extend the derived solution to address the unknown fault onset time.

5.1 Distributionally robust FD test design with a given fault onset time

To solve the chance constrained maximin optimization problem (31), the probabilistic definitions of the worst-case FAR and FDR in (31) are first transformed into equivalent deterministic expressions as follows.

Lemma 2 (Theorem 1 in [36]) *With any given matrix $\mathbf{W}_\rho^k \succeq 0$, the worst-case FAR is given by*

$$\sup_{\mathbb{P}_{\xi_\rho^k}\in\mathcal{P}(\mathbf{0},\mathbf{\Sigma}_\rho^k)} h(\mathbb{P}_{\xi_\rho^k}, \mathbf{0}, \mathbf{W}_\rho^k) = \min\{\mathrm{tr}(\mathbf{W}_\rho^k\mathbf{\Sigma}_\rho^k), 1\}. \tag{33}$$

Lemma 3 (Theorem 6.1 in [2], Section 3.1 in [28]) *With any given matrix $\mathbf{W}_\rho^k \succeq 0$, the worst-case FDR for a fault vector θ is expressed as*

$$\inf_{\mathbb{P}_{\xi_\rho^k}\in\mathcal{P}(\mathbf{0},\mathbf{\Sigma}_\rho^k)} h(\mathbb{P}_{\xi_\rho^k}, \theta, \mathbf{W}_\rho^k) = \frac{p(\theta, \mathbf{W}_\rho^k)}{1 + p(\theta, \mathbf{W}_\rho^k)}, \tag{34}$$

where $p(\theta, \mathbf{W}_\rho^k)$ is defined as

$$p(\theta, \mathbf{W}_\rho^k) = \inf_{\|r_d\|^2_{\mathbf{W}_\rho^k}\leq 1} \left\| r_d - \mathbf{F}_\rho^k\theta \right\|^2_{(\mathbf{\Sigma}_\rho^k)^{-1}}. \tag{35}$$

It should be noted that the worst-case probability bounds in Lemmas 2 and 3 can be attained or approached arbitrarily closely by distributions in the prescribed moment-based ambiguity set, where more discussions about the associated extreme distributions are referred to [2,28]. As indicated by (34) and (35), the worst-case FDR under the fault θ is a monotonically increasing function of $p(\theta, \mathbf{W}_\rho^k)$, and $p(\theta, \mathbf{W}_\rho^k)$ is the Mahalanobis distance between the fault-induced residual mean and the acceptance region. Then, the inner minimization problem in (31) is equivalent to solving $\inf_{\theta\in\Theta} p(\theta, \mathbf{W}_\rho^k)$ under the same constraint in (31). As such, the maximin optimization problem (31) can be

equivalently rewritten as

$$\begin{aligned}\max_{\mathbf{W}_\rho^k}\ \inf_{\theta,r_d}\ &\left\|r_d-\mathbf{F}_\rho^k\theta\right\|^2_{(\mathbf{\Sigma}_\rho^k)^{-1}},\\ \text{s.t. }&\mathbf{W}_\rho^k\succeq 0,\ \mathrm{tr}(\mathbf{W}_\rho^k\mathbf{\Sigma}_\rho^k)\le\gamma,\\ &\left\|\mathbf{F}_\rho^k\theta\right\|^2_{(\mathbf{\Sigma}_\rho^k)^{-1}}\ge\delta^2,\|r_d\|^2_{\mathbf{W}_\rho^k}\le 1,\end{aligned}\tag{36}$$

according to Lemmas 2 and 3 as well as (32). In the following, the above problem (36) will be tackled in two successive steps:

i) Solve the inner subproblem

$$\begin{aligned}\eta_1(\mathbf{W}_\rho^k)=\inf_{\theta,r_d}\ &\left\|r_d-\mathbf{F}_\rho^k\theta\right\|^2_{(\mathbf{\Sigma}_\rho^k)^{-1}},\\ \text{s.t. }&\left\|\mathbf{F}_\rho^k\theta\right\|^2_{(\mathbf{\Sigma}_\rho^k)^{-1}}\ge\delta^2,\|r_d\|^2_{\mathbf{W}_\rho^k}\le 1\end{aligned}\tag{37}$$

by regarding $\mathbf{W}_\rho^k$ as externally given;

ii) Using the parametric cost $\eta_1(\mathbf{W}_\rho^k)$ in (37), solve the outer subproblem

$$\max_{\mathbf{W}_\rho^k\succeq 0,\,\mathrm{tr}(\mathbf{W}_\rho^k\mathbf{\Sigma}_\rho^k)\le\gamma}\ \eta_1(\mathbf{W}_\rho^k).\tag{38}$$

For the inner problem (37), if there exists a nonzero fault signal $\tilde{\theta}\in\Theta$ such that $\|\mathbf{F}_\rho^k\tilde{\theta}\|_{\mathbf{W}_\rho^k}\le 1$ holds, we have the trivial solution $r_d=\mathbf{F}_\rho^k\tilde{\theta}$ which corresponds to a zero worst-case FDR according to (34) and (35). To avoid such a trivial solution, assume that $\|\mathbf{F}_\rho^k\theta\|_{\mathbf{W}_\rho^k}>1$ holds for all $\theta\in\Theta$. Under this condition, the non-trivial solutions to the inner problem (37) and the outer problem (38) are as follows.

Firstly, we denote the rank of $\mathbf{W}_\rho^k$ by $q=\mathrm{rank}(\mathbf{W}_\rho^k)\le n_r$, and define $\overline{\mathbf{W}}_\rho^k=(\mathbf{\Sigma}_\rho^k)^{\frac{1}{2}}\mathbf{W}_\rho^k(\mathbf{\Sigma}_\rho^k)^{\frac{1}{2}}$ whose eigenvalue decomposition is expressed as

$$\overline{\mathbf{W}}_\rho^k=\overline{U}_k\Lambda_k\overline{U}_k^\top=\begin{bmatrix}\overline{U}_{1,k}&\overline{U}_{2,k}\end{bmatrix}\mathrm{diag}(\Lambda_{1,k},\mathbf{0})\begin{bmatrix}\overline{U}_{1,k}^\top\\ \overline{U}_{2,k}^\top\end{bmatrix},\tag{39}$$

where $\overline{U}_k=\begin{bmatrix}\overline{U}_{1,k}&\overline{U}_{2,k}\end{bmatrix}\in\mathbb{R}^{n_r\times n_r}$ is an orthonormal matrix with $\overline{U}_{1,k}\in\mathbb{R}^{n_r\times q}$ and $\overline{U}_{2,k}\in\mathbb{R}^{n_r\times(n_r-q)}$, and $\Lambda_{1,k}=\mathrm{diag}(\lambda_{1,k},\lambda_{2,k},\cdots,\lambda_{q,k})$ and $\Lambda_k=\mathrm{diag}(\Lambda_{1,k},\lambda_{q+1,k},\cdots,\lambda_{n_r,k})$ are diagonal matrices with $\lambda_{1,k}\ge\lambda_{2,k}\ge\cdots\ge\lambda_{q,k}>0$ and $\lambda_{i,k}=0$ for $q<i\le n_r$.

Theorem 2 *If $\left\|\mathbf{F}_\rho^k\theta\right\|_{\mathbf{W}_\rho^k}>1$ holds for all $\theta\in\Theta$, the solution to the inner problem* (37) *is*

$$\begin{aligned}&\mathbf{F}_\rho^k\theta^*=\delta(\mathbf{\Sigma}_\rho^k)^{\frac{1}{2}}\bar{u}_{q,k},\ r_d^*=\lambda_{q,k}^{-\frac{1}{2}}(\mathbf{\Sigma}_\rho^k)^{\frac{1}{2}}\bar{u}_{q,k},\\ &\eta_1(\mathbf{W}_\rho^k)=\left(\delta-\lambda_{q,k}^{-\frac{1}{2}}\right)^2\end{aligned}\tag{40}$$

with $\delta>\lambda_{q,k}^{-\frac{1}{2}}$, where $\bar{u}_{i,k}$ denotes the ith eigenvector of $\overline{\mathbf{W}}_\rho^k$, i.e., the ith column of $\overline{U}_k$ in (39)*, and $\lambda_{q,k}>0$ is the smallest nonzero eigenvalue of $\overline{\mathbf{W}}_\rho^k$.*

The proof is given in Appendix C.

Theorem 3 *Using Theorem 2, the outer problem* (38) *becomes*

$$\begin{aligned}\max_{q,\{\lambda_{i,k}\}_{i=1}^q}\ &\left(\delta-\lambda_{q,k}^{-\frac{1}{2}}\right)^2\\ \text{s.t. }&\lambda_{1,k}\ge\lambda_{2,k}\ge\cdots\ge\lambda_{q,k}>0,\\ &\textstyle\sum_{i=1}^q\lambda_{i,k}\le\gamma, q\ge\mathrm{rank}(\mathbf{F}_\rho^k).\end{aligned}\tag{41}$$

Its optimal solution is

$$q^*=\mathrm{rank}(\mathbf{F}_\rho^k),\ \lambda_{i,k}^*=\frac{\gamma}{\mathrm{rank}(\mathbf{F}_\rho^k)},\ 1\le i\le q,\tag{42}$$

with $\delta>\sqrt{\mathrm{rank}(\mathbf{F}_\rho^k)/\gamma}$. The optima weight matrix is

$$\mathbf{W}_\rho^{k*}=\frac{\gamma(\mathbf{\Sigma}_\rho^k)^{-1}\mathbf{F}_\rho^k\left((\mathbf{F}_\rho^k)^\top(\mathbf{\Sigma}_\rho^k)^{-1}\mathbf{F}_\rho^k\right)^\dagger(\mathbf{F}_\rho^k)^\top(\mathbf{\Sigma}_\rho^k)^{-1}}{\mathrm{rank}(\mathbf{F}_\rho^k)}.\tag{43}$$

The proof is given in Appendix D.

From Theorem 3, it can be seen that we need to select $\delta>\sqrt{\mathrm{rank}(\mathbf{F}_\rho^k)/\gamma}$ for the fault set Θ defined in (32) such that the optimal solution $\mathbf{W}_\rho^{k*}$ in (43) is feasible for the problem (36). With $\mathbf{W}_\rho^{k*}$ in (43), the resulting optimal residual evaluation function (25) is

$$J^*(\mathbf{r}_\rho^k)=\left\|\mathbf{r}_\rho^k\right\|^2_{\mathbf{W}_\rho^{k*}}=\frac{\gamma\beta_{k,\rho}^\top\mathcal{H}_{k,\rho}^\dagger\beta_{k,\rho}}{\mathrm{rank}(\mathbf{F}_\rho^k)}\tag{44}$$

with

$$\begin{aligned}\beta_{k,\rho}&=(\mathbf{F}_\rho^k)^\top(\mathbf{\Sigma}_\rho^k)^{-1}\mathbf{r}_\rho^k=\sum_{i=\rho+1}^k\Phi_i^\top\left(\alpha_iR_i^*\alpha_i^\top\right)^{-1}r_i,\\ \mathcal{H}_{k,\rho}&=(\mathbf{F}_\rho^k)^\top(\mathbf{\Sigma}_\rho^k)^{-1}\mathbf{F}_\rho^k=\sum_{i=\rho+1}^k\Phi_i^\top\left(\alpha_iR_i^*\alpha_i^\top\right)^{-1}\Phi_i.\end{aligned}\tag{45}$$

The above definitions in (45) allows efficient recursive

update of $\beta_{k,\rho}$ and $\mathcal{H}_{k,\rho}$ as follows:

$$\beta_{k,\rho} = \begin{cases} \beta_{k-1,\rho} + \Phi_k^\top \left(\alpha_k R_k^* \alpha_k^\top\right)^{-1} r_k, & k > \rho+1, \\ \Phi_{\rho+1}^\top \left(\alpha_{\rho+1} R_{\rho+1}^* \alpha_{\rho+1}^\top\right)^{-1} r_{\rho+1}, & k = \rho+1, \end{cases}$$
$$\mathcal{H}_{k,\rho} = \begin{cases} \mathcal{H}_{k-1,\rho} + \Phi_k^\top \left(\alpha_k R_k^* \alpha_k^\top\right)^{-1} \Phi_k, & k > \rho+1, \\ \Phi_{\rho+1}^\top \left(\alpha_{\rho+1} R_{\rho+1}^* \alpha_{\rho+1}^\top\right)^{-1} \Phi_{\rho+1}, & k = \rho+1. \end{cases} \tag{46}$$

Remark 3 *For the sake of computation efficiency, we can update* $\{\beta_{k,\rho}, \mathcal{H}_{k,\rho}\}_{\rho=k-l+1}^{k-1}$ *by using* $l-1$ *recursions in parallel, i.e., each recursion corresponds to one possible value of* $\rho \in [k-l+1, k)$ *and follows* (46)*. When the sliding window shifts from* $[k-l, k-1)$ *to* $[k-l+1, k)$*, the oldest recursion with* $\rho = k-l$ *is deleted, the recursions initialized at* $\rho = k-1+1, \cdots, k-2$ *continue their updates according to* (46)*, and a new recursion with* $\rho = k-1$ *is added.*

5.2 Evaluation of worst-case FDR and comparison to GLR test

In this subsection, the derived FD test (44) for a fixed onset time ρ is evaluated by its worst-case FDR for any given fault θ, and compared to the well-known GLR test.

Theorem 4 *For the derived FD test* (44)*, its worst-case FDR in the presence of a specified fault* θ *is*

$$P_D^*\left(\tau_{k,\rho}(\theta), \gamma\right) = \inf_{\mathbb{P}_{\xi_\rho^k} \in \mathcal{P}(\mathbf{0}, \mathbf{\Sigma}_\rho^k)} h(\mathbb{P}_{\xi_\rho^k}, \theta, \mathbf{W}_\rho^{k*})$$
$$= \begin{cases} 0 & \text{if } \tau_{k,\rho}(\theta) \le \sqrt{\frac{\operatorname{rank}(\mathbf{F}_\rho^k)}{\gamma}}, \\ \frac{p^*(\tau_{k,\rho}(\theta),\gamma)}{1+p^*(\tau_{k,\rho}(\theta),\gamma)} & \text{if } \tau_{k,\rho}(\theta) > \sqrt{\frac{\operatorname{rank}(\mathbf{F}_\rho^k)}{\gamma}}, \end{cases} \tag{47}$$

with

$$\tau_{k,\rho}(\theta) = \left\|\mathbf{F}_\rho^k \theta\right\|_{(\mathbf{\Sigma}_\rho^k)^{-1}}, \tag{48a}$$
$$p^*(\tau_{k,\rho}(\theta), \gamma) = \left(\tau_{k,\rho}(\theta) - \sqrt{\frac{\operatorname{rank}(\mathbf{F}_\rho^k)}{\gamma}}\right)^2. \tag{48b}$$

The proof is given in Appendix E.

An interesting observation from Theorem 4 is that all faults on the ellipsoidal surface

$$\left\{\theta \,\middle|\, \tau_{k,\rho}(\theta) = \left\|\mathbf{F}_\rho^k \theta\right\|_{(\mathbf{\Sigma}_\rho^k)^{-1}} = \delta\right\}$$

gives the same worst-case FDR $P_D^*(\tau_{k,\rho}(\theta), \gamma)$ defined in (47).

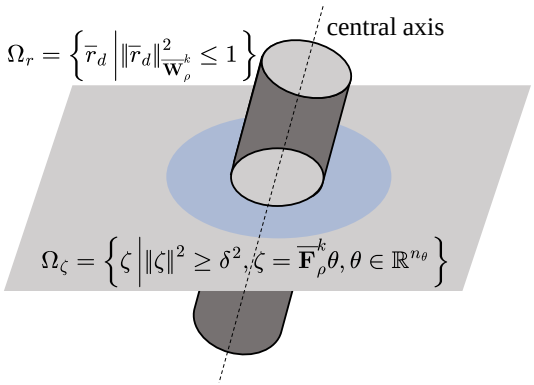


Fig. 1. Geometric interpretation of the problem (51) and its optimal solution.

It is also interesting to compare the derived FD test (44) with the well-known GLR test. For the linear residual model (26) with a zero-mean Gaussian noise $\boldsymbol{\xi}_\rho^k$ whose covariance matrix is $\mathbf{\Sigma}_k$, the GLR test is

$$(\mathbf{r}_\rho^k)^\top (\mathbf{\Sigma}_\rho^k)^{-1} \mathbf{F}_\rho^k \left((\mathbf{F}_\rho^k)^\top (\mathbf{\Sigma}_\rho^k)^{-1} \mathbf{F}_\rho^k\right)^\dagger (\mathbf{F}_\rho^k)^\top (\mathbf{\Sigma}_\rho^k)^{-1} \mathbf{r}_\rho^k$$
$$= T(\mathbf{r}_\rho^k) \begin{cases} \le T_{th}(\gamma) \Rightarrow \text{no fault alarm} \\ > T_{th}(\gamma) \Rightarrow \text{fault alarm} \end{cases} \tag{49}$$

where the detection threshold is $T_{th}(\gamma) = Q^{-1}_{\chi^2_{\operatorname{rank}(\mathbf{F}_\rho^k)}}(1-\gamma)$, with $Q^{-1}_{\chi^2_{\operatorname{rank}(\mathbf{F}_\rho^k)}}(\cdot)$ being the chi-squared inverse cumulative distribution function whose degrees of freedom is $\operatorname{rank}(\mathbf{F}_\rho^k)$. Note that our proposed test (44) can be expressed in the same form as in (49), with the detection threshold adjusted to $\operatorname{rank}(\mathbf{F}_\rho^k)/\gamma$ to compensate for the distributional ambiguity. As such, our proposed test (44) ensures the worst-case FAR to be below the desired level γ, but compromises its FDR.

Corollary 5 *For any given* $\gamma \in (0,1)$*, the worst-case FDR* $P_D^*\left(\tau_{k,\rho}(\theta), \gamma\right)$ *over the prescribed ambiguity set is lower than the nominal FDR of the GLR test under the Gaussian noise* $\boldsymbol{\xi}_\rho^k \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}_k)$*.*

The proof is given in Appendix F.

5.3 Geometric interpretation

For the problem (36) and its optimal solution derived above, a clear geometric interpretation is given as follows by introducing

$$\overline{\mathbf{W}}_\rho^k = (\mathbf{\Sigma}_\rho^k)^{\frac{1}{2}} \mathbf{W}_\rho^k (\mathbf{\Sigma}_\rho^k)^{\frac{1}{2}},$$
$$\overline{\mathbf{F}}_\rho^k = (\mathbf{\Sigma}_\rho^k)^{-\frac{1}{2}} \mathbf{F}_\rho^k, \ \overline{r}_d = (\mathbf{\Sigma}_\rho^k)^{-\frac{1}{2}} r_d, \tag{50}$$

to equivalently express (36) as

$$\max_{\overline{\mathbf{W}}_\rho^k \succeq 0, \mathrm{tr}(\overline{\mathbf{W}}_\rho^k)\le\gamma} \ \inf_{\left\|\overline{\mathbf{F}}_\rho^k\theta\right\|^2 \ge \delta^2, \|\overline{r}_d\|^2_{\overline{\mathbf{W}}_\rho^k}\le 1} \left\|\overline{r}_d - \overline{\mathbf{F}}_\rho^k\theta\right\|^2. \tag{51}$$

An illustrative example is depicted in Fig. 1, where $n_r = 3$, $n_\theta = 2$, and $\mathrm{rank}(\overline{\mathbf{W}}_\rho^k) = 2$. The inner problem in (51) computes the distance between two sets in the n_r-dimensional space. The cylinder set $\Omega_r = \left\{\overline{r}_d \,\middle|\, \|\overline{r}_d\|^2_{\overline{\mathbf{W}}_\rho^k} \le 1\right\}$ is unbounded along its central axis, and varies with $\overline{\mathbf{W}}_\rho^k$. The other set $\Omega_\zeta = \left\{\zeta \,\middle|\, \|\zeta\|^2 \ge \delta^2, \zeta = \overline{\mathbf{F}}_\rho^k\theta, \theta\in\mathbb{R}^{n_\theta}\right\}$ is restricted in the n_θ-dimensional plane $\mathrm{col}(\overline{\mathbf{F}}_\rho^k)$ (i.e., the gray plane in Fig. 1), and complementary to a 2-dimensional ball whose radii is δ, i.e., the blue ellipse in Fig. 1. According to (43), the optimal solution $\overline{\mathbf{W}}_\rho^{k*}$ to the outer problem in (51) is $\overline{\mathbf{W}}_\rho^{k*} = \frac{\gamma}{\mathrm{rank}(\overline{\mathbf{F}}_\rho^k)}\overline{\mathbf{F}}_\rho^k\left((\overline{\mathbf{F}}_\rho^k)^\top\overline{\mathbf{F}}_\rho^k\right)^\dagger(\overline{\mathbf{F}}_\rho^k)^\top$ whose column space is the plane $\mathrm{col}(\overline{\mathbf{F}}_\rho^k)$. With this optimal solution, the central axis of the cylinder Ω_r becomes perpendicular to the plane $\mathrm{col}(\overline{\mathbf{F}}_\rho^k)$, and its intersection with the plane $\mathrm{col}(\overline{\mathbf{F}}_\rho^k)$ is a 2-dimensional ball with radii $\sqrt{\mathrm{rank}(\mathbf{F}_\rho^k)/\gamma}$.

In addition, we can interpret both $J^*(\mathbf{r}_\rho^k)$ in (44) and $\tau_{k,\rho}(\theta)$ in (48) in terms of orthogonal projection. With the optimal weight $\mathbf{W}_\rho^{k*}$ in (43), $J^*(\mathbf{r}_\rho^k)$ in (44) is rewritten as

$$J^*(\mathbf{r}_\rho^k) = \frac{\gamma}{\mathrm{rank}(\mathbf{F}_\rho^k)}\left\|(\mathbf{\Sigma}_\rho^k)^{-\frac{1}{2}}\mathbf{r}_\rho^k\right\|^2_{\mathcal{M}_{k,\rho}} \tag{52}$$

with

$$\mathcal{M}_{k,\rho} = (\mathbf{\Sigma}_\rho^k)^{-\frac{1}{2}}\mathbf{F}_\rho^k\left((\mathbf{F}_\rho^k)^\top(\mathbf{\Sigma}_\rho^k)^{-1}\mathbf{F}_\rho^k\right)^\dagger(\mathbf{F}_\rho^k)^\top(\mathbf{\Sigma}_\rho^k)^{-\frac{1}{2}}. \tag{53}$$

Since $\mathcal{M}_{k,\rho}$ is symmetric and satisfies $\mathcal{M}_{k,\rho}^2 = \mathcal{M}_{k,\rho}$, $\mathcal{M}_{k,\rho}$ is a projector that projects vectors onto its column space $\mathrm{col}(\mathcal{M}_{k,\rho}) = \mathrm{col}\left((\mathbf{\Sigma}_\rho^k)^{-\frac{1}{2}}\mathbf{F}_\rho^k\right)$. As such, $J^*(\mathbf{r}_\rho^k)$ in (52) can be regarded as the result of projecting $(\mathbf{\Sigma}_\rho^k)^{-\frac{1}{2}}\mathbf{r}_\rho^k$ onto $\mathrm{col}\left((\mathbf{\Sigma}_\rho^k)^{-\frac{1}{2}}\mathbf{F}_\rho^k\right)$. Moreover, according to (26) and (48), $\tau_{k,\rho}(\theta)$ can be regarded as projecting the fault component $(\mathbf{\Sigma}_\rho^k)^{-\frac{1}{2}}\mathbf{F}_\rho^k\theta$ in $(\mathbf{\Sigma}_\rho^k)^{-\frac{1}{2}}\mathbf{r}_\rho^k$ onto $\mathrm{col}\left((\mathbf{\Sigma}_\rho^k)^{-\frac{1}{2}}\mathbf{F}_\rho^k\right)$, because

$$\left\|(\mathbf{\Sigma}_\rho^k)^{-\frac{1}{2}}\mathbf{F}_\rho^k\theta\right\|_{\mathcal{M}_{k,\rho}} = \left\|\mathbf{F}_\rho^k\theta\right\|_{(\mathbf{\Sigma}_\rho^k)^{-1}} = \tau_{k,\rho}(\theta). \tag{54}$$

5.4 Performance improvement via reduced-order fault approximation

As indicated by (47) and (48), the worst-case FDR is a monotone increasing function of $\tau_{k,\rho}(\theta) - \sqrt{\mathrm{rank}(\mathbf{F}_\rho^k)/\gamma}$ if $\tau_{k,\rho}(\theta) > \sqrt{\mathrm{rank}(\mathbf{F}_\rho^k)/\gamma}$. In this subsection, we properly introduce a reduced-order fault description $\tilde{\theta}$ to replace the original fault vector θ. The purpose is to further improve the worst-case FDR in (47) by finding a good approximation of $\tau_{k,\rho}(\theta)$ while significantly decreasing $\sqrt{\mathrm{rank}(\mathbf{F}_\rho^k)/\gamma}$, i.e., increasing $p(\theta, \mathbf{W}_\rho^{k*})$ in (48b).

To this end, we first introduce the eigenvalue decomposition of $\mathcal{H}_{k,\rho}$ in (45), i.e.,

$$\mathcal{H}_{k,\rho} = (\mathbf{F}_\rho^k)^\top(\mathbf{\Sigma}_\rho^k)^{-1}\mathbf{F}_\rho^k = V_k S_k V_k^\top \tag{55}$$

where $S_k = \mathrm{diag}(s_{1,k}, s_{2,k}, \cdots, s_{n_\theta,k})$ is a diagonal matrix with its diagonal elements in a descending order. Let the matrix $\tilde{V}_k \in \mathbb{R}^{n_\theta\times\kappa}$ consist of the first $\kappa < \theta$ columns of V_k, and define a κ-order fault approximation

$$\tilde{\theta} = \widetilde{P}_k\theta,\ \widetilde{P}_k = \widetilde{V}_k\widetilde{V}_k^\top. \tag{56}$$

By replacing θ with $\tilde{\theta}$ in all the derivations in previous sections, we obtain the following residual evaluation function $\tilde{J}(\mathbf{r}_\rho^k)$ similar to (44) and (52):

$$\tilde{J}(\mathbf{r}_\rho^k) = \frac{\gamma\tilde{\beta}_{k,\rho}^\top\widetilde{\mathcal{H}}_{k,\rho}^\dagger\tilde{\beta}_{k,\rho}}{\mathrm{rank}(\widetilde{\mathbf{F}}_\rho^k)} = \frac{\gamma\left\|(\mathbf{\Sigma}_\rho^k)^{-\frac{1}{2}}\mathbf{r}_\rho^k\right\|^2_{\widetilde{\mathcal{M}}_{k,\rho}}}{\mathrm{rank}(\widetilde{\mathbf{F}}_\rho^k)} \tag{57}$$

with

$$\widetilde{\mathbf{F}}_\rho^k = \mathbf{F}_\rho^k\widetilde{P}_k,\ \tilde{\beta}_{k,\rho} = \widetilde{P}_k\beta_{k,\rho},\ \widetilde{\mathcal{H}}_{k,\rho} = \widetilde{P}_k\mathcal{H}_{k,\rho}\widetilde{P}_k, \tag{58a}$$

$$\widetilde{\mathcal{M}}_{k,\rho} = (\mathbf{\Sigma}_\rho^k)^{-\frac{1}{2}}\widetilde{\mathbf{F}}_\rho^k\widetilde{\mathcal{H}}_{k,\rho}^\dagger(\widetilde{\mathbf{F}}_\rho^k)^\top(\mathbf{\Sigma}_\rho^k)^{-\frac{1}{2}}. \tag{58b}$$

Then, the associated worst-case FDR is

$$\widetilde{P}_D\left(\tilde{\tau}_{k,\rho}(\theta),\gamma\right) = \begin{cases} 0 & \text{if } \tilde{\tau}_{k,\rho}(\theta) \le \sqrt{\frac{\mathrm{rank}(\widetilde{\mathbf{F}}_\rho^k)}{\gamma}}, \\ \frac{\tilde{p}(\tilde{\tau}_{k,\rho}(\theta),\gamma)}{1+\tilde{p}(\tilde{\tau}_{k,\rho}(\theta),\gamma)}, & \text{if } \tilde{\tau}_{k,\rho}(\theta) > \sqrt{\frac{\mathrm{rank}(\widetilde{\mathbf{F}}_\rho^k)}{\gamma}}, \end{cases} \tag{59}$$

with

$$\tilde{\tau}_{k,\rho}(\theta) = \left\|(\mathbf{\Sigma}_\rho^k)^{-\frac{1}{2}}\mathbf{F}_\rho^k\theta\right\|_{\widetilde{\mathcal{M}}_{k,\rho}}, \tag{60}$$

$$\tilde{p}(\tilde{\tau}_{k,\rho}(\theta),\gamma) = \left(\tilde{\tau}_{k,\rho}(\theta) - \sqrt{\frac{\mathrm{rank}(\widetilde{\mathbf{F}}_\rho^k)}{\gamma}}\right)^2. \tag{61}$$

Similar to $\mathcal{M}_{k,\rho}$ in (53), $\widetilde{\mathcal{M}}_{k,\rho}$ in (58b) represents an orthogonal projection onto the column space $\mathrm{col}\left((\mathbf{\Sigma}_\rho^k)^{-\frac{1}{2}}\widetilde{\mathbf{F}}_\rho^k\right)$. According to (58a), since $\widetilde{P}_k = \widetilde{V}_k\widetilde{V}_k^\top$ is an orthogonal projection onto a reduced-order fault parameter space, the following relations hold:

$$\begin{aligned}&\mathrm{rank}(\widetilde{\mathbf{F}}_\rho^k) < \mathrm{rank}(\mathbf{F}_\rho^k),\\ &\mathrm{col}\left((\mathbf{\Sigma}_\rho^k)^{-\frac{1}{2}}\widetilde{\mathbf{F}}_\rho^k\right) \subset \mathrm{col}\left((\mathbf{\Sigma}_\rho^k)^{-\frac{1}{2}}\mathbf{F}_\rho^k\right).\end{aligned} \tag{62}$$

As such, $\tilde{\tau}_{k,\rho}(\theta)$ in (60) and $\tau_{k,\rho}(\theta)$ in (54) represent different orthogonal projections of the same fault component $(\mathbf{\Sigma}_\rho^k)^{-\frac{1}{2}}\mathbf{F}_\rho^k\theta$ onto $\mathrm{col}\left((\mathbf{\Sigma}_\rho^k)^{-\frac{1}{2}}\widetilde{\mathbf{F}}_\rho^k\right)$ and $\mathrm{col}\left((\mathbf{\Sigma}_\rho^k)^{-\frac{1}{2}}\mathbf{F}_\rho^k\right)$, respectively. Due to the second relation of the two projected spaces in (62), we have

$$\tilde{\tau}_{k,\rho}(\theta) < \tau_{k,\rho}(\theta). \tag{63}$$

As indicated in (62) and (63), both $\mathrm{rank}(\widetilde{\mathbf{F}}_\rho^k)$ and $\tilde{\tau}_{k,\rho}(\theta)$ become smaller due to introducing the reduced-order fault approximation in (56). On one hand, the term $\sqrt{\mathrm{rank}(\widetilde{\mathbf{F}}_\rho^k)/\gamma}$ could become significantly smaller, since the desired FAR level γ is often set close to zero. On the other hand, we can obtain only a slight decrease from $\tau_{k,\rho}(\theta)$ to $\tilde{\tau}_{k,\rho}(\theta)$ by properly constructing $\tilde{V}_k$ in the κ-order fault approximation (56) according to the eigenvalue decomposition (55). Specifically, we choose κ such that the sum of the κ largest eigenvalues in (55) takes up a sufficiently large portion (e.g., 95%) of $\mathrm{tr}(\mathcal{H}_{k,\rho})$, and then use the eigenvectors corresponding to these κ largest eigenvalues to construct $\widetilde{V}_k$. By using such a κ-order fault approximation, the difference $\tilde{\tau}_{k,\rho}(\theta) - \sqrt{\mathrm{rank}(\widetilde{\mathbf{F}}_\rho^k)/\gamma}$ increases, leading to an improved worst-case FDR.

5.5 Extension to unknown fault onset time

In the previous subsections, the derivations assume that the fault onset time ρ is known. To perform the FD test (23) with an unknown fault onset time ρ, all possible values of $\rho \in [k-l+1, k)$ are used to compute the residual evaluation function $J^*(\mathbf{r}_\rho^k)$ in (44), or $\widetilde{J}(\mathbf{r}_\rho^k)$ in (57) when adopting the reduced-order fault approximation (56).

In Algorithm 1, we summarize the proposed distributionally robust FD algorithm for the LTV system (1), which includes the reduced-order fault approximation. If the original full-order fault description is adopted in the FD test, we just need to modify Algorithm 1 by removing Lines 17–19 and replacing $\widetilde{J}(\mathbf{r}_\rho^k)$ in Lines 20–21 with $J^*(\mathbf{r}_\rho^k)$ in (44).

Algorithm 1 Distributionally robust FD for the LTV system (1)

1: *Initialization*: $\hat{x}_{0|0} \leftarrow \bar{x}_0$, $P_{0|0} \leftarrow P_0^x$, $k \leftarrow 1$
2: **loop**
3: ▷ *Residual generation* ◁
4: Find G_{k-1} from the full rank decomposition of E_{k-1} according to (8);
5: Compute M_k as the left inverse of C_kG_{k-1} according to (10);
6: Compute A_{k-1}^* in (12d), Q_{k-1}^* in (14a), P_k^* in (14b), S_k^* in (14c), R_k^* in (14d), and V_k^* in (14f);
7: Find α_k satisfying (16) and set $\Xi_k = \alpha_k$ as in (18);
8: Compute K_k in (17), L_k in (9), and Π_k in (11);
9: Update r_k and $\hat{x}_{k|k}$ according to (3);
10: Update $P_{k|k}$ using (13);
11: **if** $k \geq l$ **then**
12: **for** $\rho = k-l+1 \to k-1$ **do**
13: ▷ *FD test with a given ρ* ◁
14: Compute $\{\Phi_i(\rho)\}_{i=\rho+1}^k$ using (21) and (22);
15: Compute $\{\beta_{i,\rho}, \mathcal{H}_{i,\rho}\}_{i=\rho+1}^k$ using (46) and the parallel recursive algorithm in Remark 3;
16: ▷ *Reduced-order fault approximation* ◁
17: Perform eigenvalue decomposition of $\mathcal{H}_{k,\rho}$ as in (55);
18: Choose κ such that the sum of the κ largest eigenvalues of $\mathcal{H}_{k,\rho}$ takes up a sufficiently large portion (e.g., 80%) of $\mathrm{tr}(\mathcal{H}_{k,\rho})$, and then use the eigenvectors corresponding to these κ largest eigenvalues to construct $\tilde{V}_k$;
19: Compute $\tilde{\beta}_{k,\rho}$ and $\widetilde{\mathcal{H}}_{k,\rho}$ in (58a) using $\widetilde{P}_k$ in (56);
20: Compute $\tilde{J}(\mathbf{r}_\rho^k)$ according to (57);
21: **if** $\tilde{J}(\mathbf{r}_\rho^k) > 1$ **then**
22: Trigger an alarm at time k
23: **end if**
24: **end for**
25: **end if**
26: $k \leftarrow k+1$
27: **end loop**

6 Case studies

In this section, we demonstrate the effectiveness of our proposed distributionally robust design by using a numerical example and a quadruple water tank system.

6.1 *Numerical example*

Consider the LTV system in the form of (1), with

$$A_k = \begin{bmatrix} 0.0164 & 0.3339 & 0.1e^{-k} & -0.2676 & 0.1853 \\ 0.0707 & -0.1930 & 0.1\cos(k) & -0.4164 & 0.0179 \\ 0.1\sin(k) & 0.3936 & -0.3181 & 0.1035 & -0.4515 \\ -0.3285 & 0.2215 & 0.2856 & 0.2290 & -0.3621 \\ 0.1853 & -0.3101 & 0.4655 & -0.2238 & -0.3130 \end{bmatrix},$$

$$B_k = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \; E_k = \begin{bmatrix} -1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \; C_k = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

The unknown input signal is $d_k = 20\sin(0.1k) + 10$.

In the simulation example, both noises w_k and v_k follow a Gaussian mixture distribution, i.e.,

$$\star_k \sim (1-\pi_\star)\mathcal{N}(\mu_{\star,1}, \Sigma_{\star,1}) + \pi_\star \mathcal{N}(\mu_{\star,2}, \Sigma_{\star,2}), \quad (64)$$

where $\star$ represents w or v, $\pi_w = 0.25$, $\mu_{w,1} = -0.5\mathbb{1}_5$, $\Sigma_{w,1} = 0.2667 I_5$, $\mu_{w,2} = 1.5\mathbb{1}_5$, $\Sigma_{w,2} = 0.2 I_5$, $\pi_v = 0.05$, $\mu_{v,1} = -0.0789\mathbb{1}_4$, $\Sigma_{v,1} = 0.9175 I_4$, $\mu_{v,2} = 1.5\mathbb{1}_4$, and $\Sigma_{v,2} = 0.2 I_4$. According to the above distributions, the stochastic noises w_k and v_k have zero mean, and their covariance matrices are $Q_k = I_5$ and $R_k = I_4$, respectively. Except for the above mean and covariance information, the specific noise distributions are unknown to the FD algorithms.

A step fault is considered in this example. In (1), the fault vector is $\theta = \mathbb{1}_5$. As described in (2), the simulated fault onset time is $\rho_* = 101$, and the fault profile matrix $\Psi_k(\rho_*)$ is defined with

$$\widetilde{\Psi}_k = \begin{bmatrix} -0.1660 & -0.8153 & -0.1616 & 0.3409 & 0.6015 \\ 0.4406 & -0.6275 & 0.3704 & -0.1654 & 0.9365 \\ -0.9998 & -0.3089 & -0.5911 & 0.1174 & -0.3732 \\ -0.3953 & -0.2065 & 0.7562 & -0.7192 & 0.3846 \\ -0.7065 & 0.0776 & -0.9452 & -0.6038 & 0.7528 \end{bmatrix}.$$

Two FD methods are implemented for comparisons: i) the GLR test in (49); and ii) our proposed distributionally robust design described in Algorithm 1. In the latter approach, the κ-order fault approximation in (55)–(58) is adopted such that the sum of the κ largest eigenvalues of $\mathcal{H}_{k,\rho}$ takes up 60% of $\mathrm{tr}(\mathcal{H}_{k,\rho})$.

With $\gamma = 0.05$ and the search window length $l = 30$, the detection results of one simulation run are depicted in Fig. 2. It can be seen from the upper subfigure of Fig. 2 that the injected step fault has an almost invisible effect on the generated residual. Both our proposed method and the GLR test can successfully trigger alarms after fault occurrence. Compared to our proposed method, the GLR test gives larger values in the residual evaluation result and a smaller detection delay. However, due to not accounting for the ambiguous distribution information, the GLR test leads to a significant number of false alarms. In contrast, our proposed method gives no false alarm in the lower subfigure of Fig. 2.

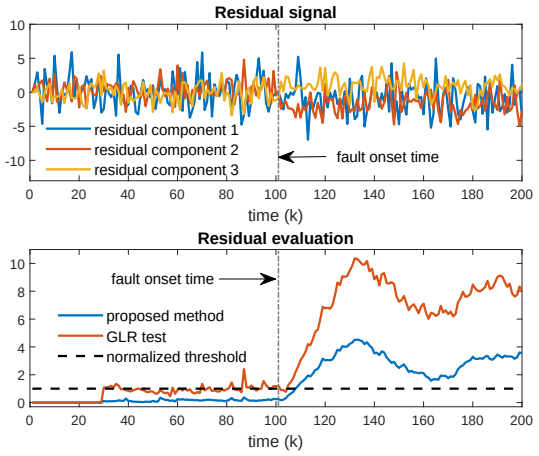


Fig. 2. Detection results for the step fault occurring at time $\rho_* = 101$ with $\theta = \mathbb{1}_5$. For our proposed method and the GLR test, the sliding window size for the unknown fault onset time ρ is $l = 30$, and $\gamma = 0.05$.

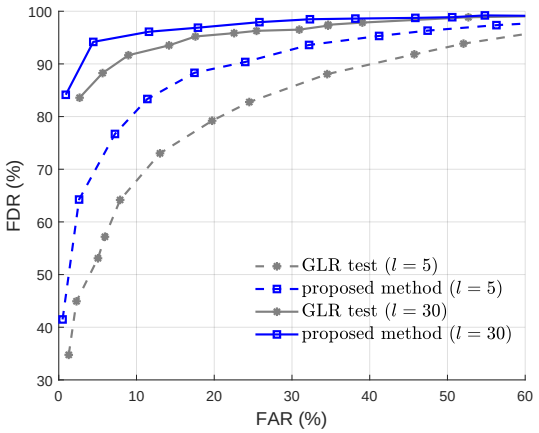


Fig. 3. ROC curves of our proposed method and the GLR test with different window lengths l for the numerical example.

We further evaluate the FAR-FDR trade-off in terms of receiver operating characteristic (ROC). The ROC curve plots the achieved FDR versus FAR. A larger area underneath the ROC curve indicates a better performance trade-off between FDR and FAR. To plot the ROC curve, we use different values of γ, and implement 100 Monte Carlo runs for each value of γ, where each run includes 200 time instants. As depicted in Fig. 3, our proposed distributionally robust design achieves a superior FAR-FDR trade-off than the GLR test. Moreover, increasing the sliding window length from $l = 5$ to $l = 30$ results in improved performance for both GLR test and our proposed DRD-FA approach, which is due to leveraging more fault information in the sliding window.

Table 1
Parameters of the quadruple water tank system.

Parameter	Notation	Value
Cross section of Tank i	Z_1, Z_2	50 cm^2
	Z_3, Z_4	28 cm^2
Cross section of outlet pipes	a_1	0.16 cm^2
	a_2	0.20 cm^2
	a_3	0.12 cm^2
	a_4	0.10 cm^2
Splitting valve opening	σ_1, σ_2	0.4
Gravity acceleration	g	981 cm/s^2

6.2 Quadruple water tank system

Consider a quadruple water tank system whose continuous-time dynamics is described as [27,32]

$$\begin{aligned}
\frac{dh_1}{dt} &= -\frac{(a_1+\theta_{a_1})\sqrt{2gh_1}}{Z_1} + \frac{(a_3+\theta_{a_3})\sqrt{2gh_3}}{Z_1} \\
&\quad + \frac{(1-\theta_{u_1})\sigma_1 u_1}{Z_1} \\
\frac{dh_2}{dt} &= -\frac{(a_2+\theta_{a_2})\sqrt{2gh_2}}{Z_2} + \frac{(a_4+\theta_{a_4})\sqrt{2gh_4}}{Z_2} \\
&\quad + \frac{(1-\theta_{u_2})\sigma_2 u_2}{Z_2} \\
\frac{dh_3}{dt} &= -\frac{(a_3+\theta_{a_3})\sqrt{2gh_3}}{Z_3} + \frac{(1-\theta_{u_2})(1-\sigma_2)u_2}{Z_3} \\
\frac{dh_4}{dt} &= -\frac{(a_4+\theta_{a_4})\sqrt{2gh_4}}{Z_4} + \frac{(1-\theta_{u_1})(1-\sigma_1)u_1}{Z_4}
\end{aligned} \tag{65}$$

where h_i (Unit: cm) represents the water level of Tank i, and u_1 and u_2 (Unit: cm^3/s) are the input flow rates controlled by two pumps. The system model parameters are listed in Table 6.2.

The system model (65) also describes latent faults including loss of control effectiveness $\theta_{u_i} \in [0,1]$ of the two pumps as well as the tank leakage faults θ_{a_i} (Unit: cm^2). We consider the loss of control effectiveness of Pump 1, i.e., θ_{u_1}, as a nuisance fault to be decoupled. The simulated fault scenarios are as follows:

$$\begin{aligned}
&\text{Unknown input: } d_k = \theta_{u_1} = 0.1, \text{ for } k \geq 0, \\
&\text{Fault scenario 1: } \theta = \begin{bmatrix} 0 & 0 & 0 & 0.0048 & 0 \end{bmatrix}^\top, \text{ for } k > 101, \\
&\text{Fault scenario 2: } \theta = \begin{bmatrix} 0 & 0 & 0 & 0 & 0.01 \end{bmatrix}^\top, \text{ for } k > 101,
\end{aligned} \tag{66}$$

with θ defined in (67).

By applying the forward Euler method with the sampling interval $t_s = 1$s, the continuous-time nonlinear system model (65) is approximated as the following discrete-time linear parameter varying (LPV) model in the form of (1):

$$\begin{aligned}
&x_k = \begin{bmatrix} h_{1,k} & h_{2,k} & h_{3,k} & h_{4,k} \end{bmatrix}^\top, \; u_k = \begin{bmatrix} u_{1,k} & u_{2,k} \end{bmatrix}^\top, \\
&d_k = \theta_{u_1}, \; \theta = \begin{bmatrix} \theta_{u_2} & \theta_{a_1} & \theta_{a_2} & \theta_{a_3} & \theta_{a_4} \end{bmatrix}^\top, \\
&A_k = t_s \begin{bmatrix} -\frac{2ga_1}{Z_1 c_{1,k}} & 0 & \frac{2ga_3}{Z_1 c_{3,k}} & 0 \\ 0 & -\frac{2ga_2}{Z_2 c_{2,k}} & 0 & \frac{2ga_4}{Z_2 c_{4,k}} \\ 0 & 0 & -\frac{2ga_3}{Z_3 c_{3,k}} & 0 \\ 0 & 0 & 0 & -\frac{2ga_4}{Z_4 c_{4,k}} \end{bmatrix} + I_4, \\
&B_k = \begin{bmatrix} \frac{\sigma_1}{Z_1} & 0 \\ 0 & \frac{\sigma_2}{Z_2} \\ 0 & \frac{1-\sigma_2}{Z_3} \\ \frac{1-\sigma_1}{Z_4} & 0 \end{bmatrix}, \; E_k = -\begin{bmatrix} \frac{\sigma_1 u_{1,k}}{Z_1} \\ 0 \\ 0 \\ \frac{(1-\sigma_1)u_{1,k}}{Z_4} \end{bmatrix}, \; C_k = I_4, \\
&\Psi_k = \begin{bmatrix} 0 & -\frac{c_{1,k}}{Z_1} & 0 & \frac{c_{3,k}}{Z_1} & 0 \\ -\frac{\sigma_2 u_{2,k}}{Z_2} & 0 & -\frac{c_{2,k}}{Z_2} & 0 & \frac{c_{4,k}}{Z_2} \\ -\frac{(1-\sigma_2)u_{2,k}}{Z_3} & 0 & 0 & \frac{c_{3,k}}{Z_3} & 0 \\ 0 & 0 & 0 & 0 & -\frac{c_{4,k}}{Z_4} \end{bmatrix}, \text{ for } k \geq \rho, \\
&\Psi_k = 0, \text{ for } k < \rho,
\end{aligned} \tag{67}$$

with $c_{i,k} = \sqrt{2gh_{i,k}}$.

The quadruple water tank system is controlled by the passivity-based nonlinear model predictive control proposed in [32]. The water levels vary from 1cm to 28cm, which implies the necessity of using the above LPV model (67). As in the numerical example of Section 6.1, process and measurement noises with time-varying Gaussian mixture distributions in (64) are included. Note that the specific probability distributions of noises are unknown except for their covariance matrices $Q_k = 4 \times 10^{-4} I_4$ and $R_k = 0.01 I_4$. In the above setting, we expect the detector to decouple the residual signal from the loss of control effectiveness of Pump 1, and to detect the leakages of Tank 3 and Tank 4.

For comparison, the conventional GLR test, the parity space vector machine (PSVM) in [49], and our proposed distributionally robust design are implemented. In our approach, the κ-order fault approximation in (55)–(58) is adopted such that the sum of the κ largest eigenvalues of $\mathcal{H}_{k,\rho}$ takes up 80% of $\mathrm{tr}(\mathcal{H}_{k,\rho})$. Since the original PSVM approach in [49] considered linear time-invariant systems without unknown inputs, we adopt it here by recalculating the party vector at each time instant to accommodate the time-varying dynamics and decouple the unknown input.

With $\gamma = 0.2$ and the sliding window length $l = 10$, the

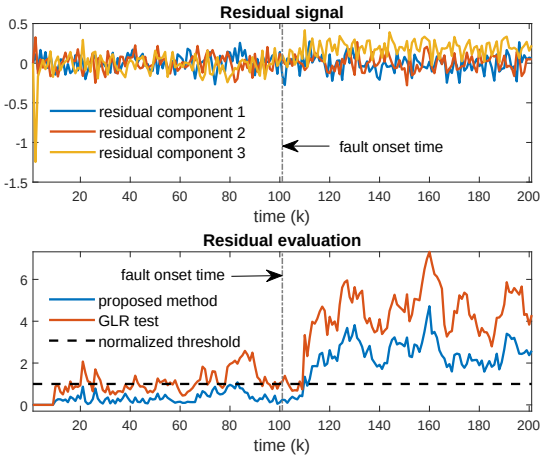


Fig. 4. Detection results for the fault scenario 1 in (66). For our proposed method and the GLR test, the sliding window size for the unknown fault onset time ρ is $l = 10$, and $\gamma = 0.2$.

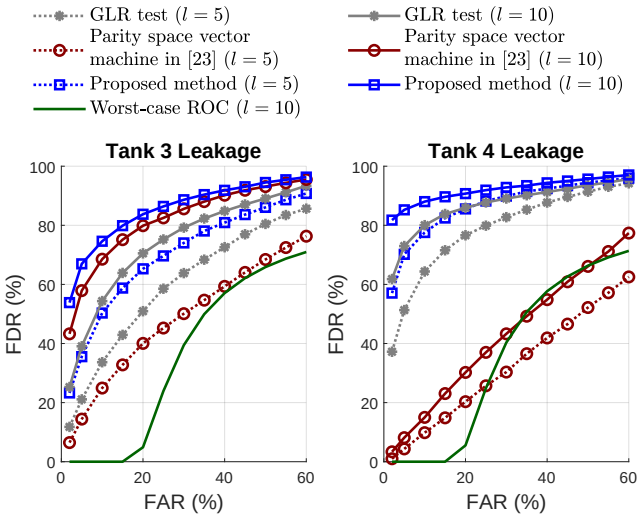


Fig. 5. ROC curves of our proposed method, the GLR test, and the parity space vector machine in [49] with different window lengths l for the quadruple water tank system.

detection results of one simulation run are given in Fig. 4. Although both our proposed method and the GLR test achieve almost the same FDR, our proposed method gives substantially smaller FAR than the GLR test. To fully evaluate the FAR-FDR trade-off, the receiver operating characteristic (ROC) curves are plotted by running Monte Carlo simulations, as shown in Fig. 5. Each data point in the plotted ROC curve is obtained from 100 Monte Carlo runs. As expected, all methods achieve low FDRs for the small leakage in Tank 3 (see Scenario 1 in (66)), while their ROC curves improve with the larger leakage in Tank 4 (see Scenario 2 in (66)) and an increased length l of the detection window. In both two fault scenarios, our proposed distributionally robust test achieves a superior FAR-FDR trade-off compared with the conventional GLR test by explicitly accounting for distributional ambiguity. Since the empirical distribution used in the simulation differs from the worst-case distribution, the empirical ROC of our proposed test exhibits higher FDRs than its worst-case ROC. The PSVM approach in [49] gives the poorest ROC performance in Fig. 5, in some cases even underperforming the conventional GLR test. This may be attributed to its use of a scalar residual, which is less effective for detecting multi-dimensional fault vectors than using a residual vector in the GLR test and our proposed method.

7 Conclusions

In this paper, we present a distributionally robust approach to FD for stochastic linear time-varying systems that are affected by both inexact noise distributions and non-random unknown inputs. Under moment-based distributional ambiguity, the worst-case FDR is maximized, while ensuring a desired level of worst-case FAR. The derived FD test is in the same form as the classical GLR test, and compensates for distributional ambiguity with a modified threshold. The obtained FD test can be efficiently computed via a parallel recursive algorithm. By analyzing the closed-form expression of the worst-case FDR, the use of a low-order fault approximation is motivated to further enhance the FDR. Future work would be devoted to incorporating more distribution information such as bounded support or higher-order moments for improved FAR-FDR trade-off.

A Proof of Theorem 1

Using (12a)–(12c), $r_{s,k+i}$ can be expressed as

$$r_{s,k+i} = \Upsilon_k^x \tilde{x}_{s,k|k} + \sum_{j=k}^{k+i-1} \Upsilon_j^w w_j^* + \Upsilon_{j+1}^v v_{j+1},\ i > 0$$

where the coefficient matrices Υ_k^x and $\{\Upsilon_j^w, \Upsilon_{j+1}^v\}_{j=k}^{k+i-1}$ are determined by the system matrices in (12a)–(12c). Since $\{w_j^*, v_{j+1}\}_{j=k}^{k+i-1}$ are all uncorrelated with $r_{s,k}$, we have $\mathbb{E}[r_{s,k} r_{s,k+i}^\top] = \mathbb{E}[r_{s,k} \tilde{x}_{s,k|k}^\top](\Upsilon_k^x)^\top$ for $i > 0$. Therefore, we can prove $\mathbb{E}[r_{s,k} r_{s,k+i}^\top] = 0$ for $i > 0$ by showing $\mathbb{E}[r_{s,k} \tilde{x}_{s,k|k}^\top] = 0$ as follows. According to (12a), (14), and (17), we express $\mathbb{E}[r_{s,k} r_{s,k+i}^\top]$ $(i > 0)$ as

$$\begin{aligned}
&\mathbb{E}[r_{s,k} r_{s,k+i}^\top] = \mathbb{E}[r_{s,k} \tilde{x}_{s,k|k}^\top](\Upsilon_k^x)^\top \\
&= \Xi_k \mathbb{E}\left\{(C_k z_{s,k} + v_k)\left[(I_{n_x} - K_k C_k) z_{s,k} - K_k v_k\right]^\top\right\}(\Upsilon_k^x)^\top \\
&= \Xi_k \left[V_k^{*\top}(I_{n_x} - K_k C_k)^\top - C_k S_k^* K_k^\top - R_k K_k^\top\right](\Upsilon_k^x)^\top \\
&= \Xi_k \left[V_k^{*\top} - R_k^* K_k^\top\right](\Upsilon_k^x)^\top \\
&= \alpha_k \left[V_k^{*\top} - R_k^* \alpha_k^\top (\alpha_k R_k^* \alpha_k^\top)^{-1} \alpha_k V_k^{*\top}\right](\Upsilon_k^x)^\top = 0,
\end{aligned}$$

where the last row leverages (17) and (18). Note that $\mathbb{E}[r_{s,k} r_{s,k+i}^\top] = 0$ $(i > 0)$ cannot be achieved with an

arbitrarily chosen Ξ_k. In addition, (19) is obtained by using (12b), (14d) and (18).

B Proof of Proposition 1

By substituting (9), (10), (11) and (18) into (4), the error dynamics in the faulty case can be rewritten as

$$\begin{aligned}\tilde{x}_{k|k} &= (I_{n_x} - K_k C_k)\left[A^\star_{k-1}\tilde{x}_{k-1|k-1} + w^\star_{k-1} \right.\\ &\quad \left. +\Psi^*_{k-1}(\rho)\theta\right] - K_k v_k, \qquad \text{(B.1a)}\\ r_k &= \alpha_k\left[C_k(A^*_{k-1}\tilde{x}_{k-1|k-1} + \Psi^*_{k-1}\theta + w^*_{k-1}) + v_k\right] \qquad \text{(B.1b)}\end{aligned}$$

with $\Psi^*_{k-1}(\rho)$ defined in (22). Using (B.1a) and (20a), we have

$$\begin{aligned}&\tilde{x}_{s,k|k} = \tilde{x}_{k|k} - \Gamma_k(\rho)\theta\\ &= (I_{n_x} - K_k C_k)\left[A^\star_{k-1}(\tilde{x}_{s,k-1|k-1} + \Gamma_{k-1}(\rho)\theta) + w^\star_{k-1}\right.\\ &\quad \left. +\Psi^\star_{k-1}(\rho)\theta\right] - K_k v_k - \Gamma_k(\rho)\theta\\ &= (I_{n_x} - K_k C_k)\left(A^\star_{k-1}\tilde{x}_{s,k-1|k-1} + w^\star_{k-1}\right) - K_k v_k\\ &\quad + \left[(I_{n_x} - K_k C_k)(A^\star_{k-1}\Gamma_{k-1}(\rho) + \Psi^\star_{k-1}(\rho)) - \Gamma_k(\rho)\right]\theta. \qquad \text{(B.2)}\end{aligned}$$

By satisfying (21a) to let the last term in (B.2) vanish, (B.2) becomes exactly the same as the fault-free error dynamics described in (4) and (12c), as expected. Using (12b), (12c), and (20a), r_k in (B.1b) can be expressed as

$$r_k = r_{s,k} + \alpha_k C_k(A^*_{k-1}\Gamma_{k-1}(\rho) + \Psi^*_{k-1}(\rho))\theta, \qquad \text{(B.3)}$$

where the coefficient matrix before θ is exactly $\Phi_k(\rho)$ defined in (21b).

C Proof of Theorem 2

Since $\left\|\mathbf{F}^k_\rho\theta\right\|_{\mathbf{W}^k_\rho} > 1$ holds for all $\theta \in \Theta$, the inner problem (37) achieves its minimal value when $\|r_d\|^2_{\mathbf{W}^k_\rho} = 1$ and $\left\|\mathbf{F}^k_\rho\theta\right\|^2_{(\mathbf{\Sigma}^k_\rho)^{-1}} = \delta^2$. Then, the Lagrangian function for the inner problem (37) is

$$\begin{aligned}\mathcal{L}(r_d,\theta,\mu_1,\mu_2) &= \left\|r_d - \mathbf{F}^k_\rho\theta\right\|^2_{(\mathbf{\Sigma}^k_\rho)^{-1}} + \mu_1\left(\|r_d\|^2_{\mathbf{W}^k_\rho} - 1\right)\\ &\quad + \mu_2\left(\delta^2 - \left\|\mathbf{F}^k_\rho\theta\right\|^2_{(\mathbf{\Sigma}^k_\rho)^{-1}}\right) \qquad \text{(C.1)}\end{aligned}$$

with Lagrangian multipliers μ_1 and μ_2. The corresponding optimality conditions are

$$\begin{aligned}&\frac{\partial\mathcal{L}}{\partial r_d} = 2\left((\mathbf{\Sigma}^k_\rho)^{-1} + \mu_1\mathbf{W}^k_\rho\right)r_d - 2(\mathbf{\Sigma}^k_\rho)^{-1}\mathbf{F}^k_\rho\theta = \mathbf{0}, \qquad \text{(C.2a)}\\ &\frac{\partial\mathcal{L}}{\partial\theta} = 2(\mathbf{F}^k_\rho)^\top(\mathbf{\Sigma}^k_\rho)^{-1}\left((1-\mu_2)\mathbf{F}^k_\rho\theta - r_d\right) = \mathbf{0}, \qquad \text{(C.2b)}\\ &\|r_d\|^2_{\mathbf{W}^k_\rho} = 1,\ \left\|\mathbf{F}^k_\rho\theta\right\|^2_{(\mathbf{\Sigma}^k_\rho)^{-1}} = \delta^2. \qquad \text{(C.2c)}\end{aligned}$$

From (C.2a) and (C.2b), we have

$$\begin{aligned}&r_d = (\mathbf{\Sigma}^k_\rho)^{\frac{1}{2}}(I_{n_r} + \mu_1\overline{\mathbf{W}}^k_\rho)^{-1}(\mathbf{\Sigma}^k_\rho)^{-\frac{1}{2}}\mathbf{F}^k_\rho\theta, \qquad \text{(C.3)}\\ &\left[(1-\mu_2)I_{n_r} - \left(I_{n_r} + \mu_1\overline{\mathbf{W}}^k_\rho\right)^{-1}\right](\mathbf{\Sigma}^k_\rho)^{-\frac{1}{2}}\mathbf{F}^k_\rho\theta = \mathbf{0}, \qquad \text{(C.4)}\end{aligned}$$

with $\overline{\mathbf{W}}^k_\rho = (\mathbf{\Sigma}^k_\rho)^{\frac{1}{2}}\mathbf{W}^k_\rho(\mathbf{\Sigma}^k_\rho)^{\frac{1}{2}}$. Using the eigenvalue decomposition in (39), we have

$$\begin{aligned}&(1-\mu_2)I_{n_r} - \left(I_{n_r} + \mu_1\overline{\mathbf{W}}^k_\rho\right)^{-1}\\ &\quad = \overline{U}_k\left[(1-\mu_2)I_{n_r} - (I_{n_r} + \mu_1\Lambda_k)^{-1}\right]\overline{U}_k^\top\end{aligned}$$

which implies that its eigenvalues are $\{1-\mu_2-(1+\mu_1\lambda_{i,k})^{-1}\}_{i=1}^{n_r}$. To ensure the feasibility of (C.4), at least one of these eigenvalues is zero. Considering this fact, let $(\lambda_{j,k}, \bar{u}_{j,k})$ denote the pair of eigenvalue and eigenvector of $\overline{\mathbf{W}}^k_\rho$, and $\lambda_{j,k}$ satisfies $1-\mu_2-(1+\mu_1\lambda_{j,k})^{-1} = 0$. Therefore, it follows from (C.4) and the second equation in (C.2c) that $(\mathbf{\Sigma}^k_\rho)^{-\frac{1}{2}}\mathbf{F}^k_\rho\theta = \delta\bar{u}_{j,k}$ holds. Below we consider two scenarios.

Scenario 1. If $1-\mu_2-(1+\mu_1\lambda_{j,k})^{-1} = 0$ holds for $\lambda_{j,k} = 0$, $q < j \le n_r$, r_d in (C.3) becomes $r_d = (\mathbf{\Sigma}^k_\rho)^{\frac{1}{2}}\delta\bar{u}_{j,k}$ using $(\mathbf{\Sigma}^k_\rho)^{-\frac{1}{2}}\mathbf{F}^k_\rho\theta = \delta\bar{u}_{j,k}$ and (39). In this case, we have $\|r_d\|^2_{\mathbf{W}^k_\rho} = \|\delta\bar{u}_{j,k}\|^2_{\overline{\mathbf{W}}^k_\rho} = 0$ which is in contradiction to the first equation in (C.2c).

Scenario 2. If $1-\mu_2-(1+\mu_1\lambda_{j,k})^{-1} = 0$ holds for $\lambda_{j,k} > 0$, $1 \le i \le q$, r_d in (C.3) becomes $r_d = (\mathbf{\Sigma}^k_\rho)^{\frac{1}{2}}\frac{\delta\bar{u}_{j,k}}{1+\mu_1\lambda_{j,k}}$. Then, we have $\|r_d\|^2_{\mathbf{W}^k_\rho} = \left\|\frac{\delta\bar{u}_{j,k}}{1+\mu_1\lambda_{j,k}}\right\|^2_{\overline{\mathbf{W}}^k_\rho} = \frac{\delta^2\lambda_{j,k}}{(1+\mu_1\lambda_{j,k})^2} = 1$ which implies $1+\mu_1\lambda_{j,k} = \delta\lambda^{\frac{1}{2}}_{j,k}$. In this case, the solution is $r_d = \lambda^{-\frac{1}{2}}_{j,k}\mathbf{\Sigma}^{\frac{1}{2}}_k\bar{u}_{j,k}$, and the corresponding cost function can be expressed as $\left\|r_d - \mathbf{F}^k_\rho\theta\right\|^2_{(\mathbf{\Sigma}^k_\rho)^{-1}} = \left\|\left(\lambda^{-\frac{1}{2}}_{j,k} - \delta\right)\bar{u}_{j,k}\right\|^2 = \left(\delta - \lambda^{-\frac{1}{2}}_{j,k}\right)^2$. Therefore, the inner problem (37) is equivalent to $\min_j\ (\delta - \lambda^{-\frac{1}{2}}_{j,k})^2$, and it achieves its minimum with the smallest eigenvalue $\lambda_{j^*,k} = \lambda_{q,k}$, which proves the optimal solution in (40).

The feasibility of the solution θ^* in (40) requires $\left\|\mathbf{F}^k_\rho\theta^*\right\|_{\mathbf{W}^k_\rho} > 1$ which implies $\|\delta\bar{u}_{q,k}\|_{\overline{\mathbf{W}}^k_\rho} = \delta\lambda^{\frac{1}{2}}_{q,k} > 1$.

D Proof of Theorem 3

The condition $\left\|\mathbf{F}^k_\rho\theta\right\|_{\mathbf{W}^k_\rho} = \left\|(\mathbf{\Sigma}^k_\rho)^{-\frac{1}{2}}\mathbf{F}^k_\rho\theta\right\|_{\overline{\mathbf{W}}^k_\rho} > 1$ for all $\theta \in \Theta$ implies that there does not exist any

$(\boldsymbol{\Sigma}_\rho^k)^{-\frac{1}{2}}\mathbf{F}_\rho^k\theta \neq 0$ lying in the nullspace of $\overline{\mathbf{W}}_\rho^k$, i.e.,

$$\begin{aligned}&\operatorname{col}\left(\overline{\mathbf{W}}_\rho^k\right) \supseteq \operatorname{col}\left((\boldsymbol{\Sigma}_\rho^k)^{-\frac{1}{2}}\mathbf{F}_\rho^k\right),\\ &\operatorname{rank}\left(\overline{\mathbf{W}}_\rho^k\right) = q \geq \operatorname{rank}\left((\boldsymbol{\Sigma}_\rho^k)^{-\frac{1}{2}}\mathbf{F}_\rho^k\right) = \operatorname{rank}\left(\mathbf{F}_\rho^k\right).\end{aligned} \tag{D.1}$$

Together with (40) and

$$\begin{aligned}\sum_{i=1}^{q}\lambda_{i,k} &= \operatorname{tr}\left(\overline{\mathbf{W}}_\rho^k\right) = \operatorname{tr}\left((\boldsymbol{\Sigma}_\rho^k)^{\frac{1}{2}}\mathbf{W}_\rho^k(\boldsymbol{\Sigma}_\rho^k)^{\frac{1}{2}}\right)\\ &= \operatorname{tr}\left(\mathbf{W}_\rho^k\boldsymbol{\Sigma}_\rho^k\right) \leq \gamma,\end{aligned}$$

the outer problem (38) can be equivalently expressed as (41). Due to $\delta > \lambda_{q,k}^{-\frac{1}{2}}$ derived in Theorem 2, the problem (41) achieves its maximum when $\lambda_{q,k}$ takes its maximal value. Since $\sum_{i=1}^{q}\lambda_{i,k} \leq \gamma$, $\lambda_{1,k} \geq \cdots \geq \lambda_{q,k} > 0$ and $q \geq \operatorname{rank}(\mathbf{F}_\rho^k)$ in (D.1), it is easy to see that the maximal value of $\lambda_{q,k}$ is achieved with (42). The condition $\delta > \lambda_{q,k}^{-\frac{1}{2}}$ derived in Theorem 2 becomes $\delta > \sqrt{\operatorname{rank}(\mathbf{F}_\rho^k)/\gamma}$ according to (42).

For the eigenvalue decomposition in (39), let $\overline{\mathbf{W}}_\rho^{k*}$ and $\overline{U}_{1,k}^*$ denote the optimal choice of $\overline{\mathbf{W}}_\rho^k$ and $\overline{U}_{1,k}$, respectively. From (40), (42) and (D.1), we have $\operatorname{col}(\overline{U}_{1,k}^*) = \operatorname{col}\left(\overline{\mathbf{W}}_\rho^{k*}\right) = \operatorname{col}\left((\boldsymbol{\Sigma}_\rho^k)^{-\frac{1}{2}}\mathbf{F}_\rho^k\right)$. This further implies

$$\begin{aligned}&\overline{U}_{1,k}^*(\overline{U}_{1,k}^*)^\top =\\ &\quad(\boldsymbol{\Sigma}_\rho^k)^{-\frac{1}{2}}\mathbf{F}_\rho^k\left((\mathbf{F}_\rho^k)^\top(\boldsymbol{\Sigma}_\rho^k)^{-1}\mathbf{F}_\rho^k\right)^\dagger(\mathbf{F}_\rho^k)^\top(\boldsymbol{\Sigma}_\rho^k)^{-\frac{1}{2}},\end{aligned} \tag{D.2}$$

because $\overline{U}_{1,k}^*(\overline{U}_{1,k}^*)^\top$ represents an orthogonal projection onto $\operatorname{col}\left((\boldsymbol{\Sigma}_\rho^k)^{-\frac{1}{2}}\mathbf{F}_\rho^k\right)$. Therefore, we have

$$\begin{aligned}\overline{\mathbf{W}}_\rho^{k*} &= \lambda_{i,k}^*\overline{U}_{1,k}^*(\overline{U}_{1,k}^*)^\top\\ &= \frac{\gamma(\boldsymbol{\Sigma}_\rho^k)^{-\frac{1}{2}}\mathbf{F}_\rho^k\left((\mathbf{F}_\rho^k)^\top(\boldsymbol{\Sigma}_\rho^k)^{-1}\mathbf{F}_\rho^k\right)^\dagger(\mathbf{F}_\rho^k)^\top(\boldsymbol{\Sigma}_\rho^k)^{-\frac{1}{2}}}{\operatorname{rank}(\mathbf{F}_\rho^k)},\end{aligned} \tag{D.3}$$

then (43) is derived according to $\overline{\mathbf{W}}_\rho^{k*} = (\boldsymbol{\Sigma}_\rho^k)^{\frac{1}{2}}\mathbf{W}_\rho^{k*}(\boldsymbol{\Sigma}_\rho^k)^{\frac{1}{2}}$.

E Proof of Theorem 4

According to Lemma 3, the worst-case FDR evaluation requires to compute $p(\theta, \mathbf{W}_\rho^{k*})$ by solving the problem (35) with $\mathbf{W}_k = \mathbf{W}_\rho^{k*}$. By substituting (43) into $p(\theta, \mathbf{W}_\rho^{k*})$ in (35), we have

$$p(\theta, \mathbf{W}_\rho^{k*})\begin{cases} = 0 \text{ if } \left\|\mathbf{F}_\rho^k\theta\right\|^2_{\mathbf{W}_\rho^{k*}} = \frac{\gamma\left\|\mathbf{F}_\rho^k\theta\right\|^2_{(\boldsymbol{\Sigma}_\rho^k)^{-1}}}{\operatorname{rank}(\mathbf{F}_\rho^k)} \leq 1,\\ > 0 \text{ if } \left\|\mathbf{F}_\rho^k\theta\right\|^2_{\mathbf{W}_\rho^{k*}} = \frac{\gamma\left\|\mathbf{F}_\rho^k\theta\right\|^2_{(\boldsymbol{\Sigma}_\rho^k)^{-1}}}{\operatorname{rank}(\mathbf{F}_\rho^k)} > 1.\end{cases}$$

Hence, we have (47) according to (34) and (48a). Next, we compute the nonzero $p(\theta, \mathbf{W}_\rho^{k*})$ under the condition $\left\|\mathbf{F}_\rho^k\theta\right\|^2_{\mathbf{W}_\rho^{k*}} = \frac{\gamma}{\operatorname{rank}(\mathbf{F}_\rho^k)}\left\|\mathbf{F}_\rho^k\theta\right\|^2_{(\boldsymbol{\Sigma}_\rho^k)^{-1}} > 1$. Similarly to (C.1)–(C.3), the optimality condition for the problem (35) is expressed as

$$(\boldsymbol{\Sigma}_\rho^k)^{-\frac{1}{2}}r_d^* = (I_{n_r} + \nu\overline{\mathbf{W}}_\rho^{k*})^{-1}(\boldsymbol{\Sigma}_\rho^k)^{-\frac{1}{2}}\mathbf{F}_\rho^k\theta, \tag{E.1a}$$

$$\|r_d^*\|^2_{\mathbf{W}_\rho^{k*}} = \left\|(\boldsymbol{\Sigma}_\rho^k)^{-\frac{1}{2}}r_d^*\right\|^2_{\overline{\mathbf{W}}_\rho^{k*}} = 1, \tag{E.1b}$$

where $\overline{\mathbf{W}}_\rho^{k*} = (\boldsymbol{\Sigma}_\rho^k)^{\frac{1}{2}}\mathbf{W}_\rho^{k*}(\boldsymbol{\Sigma}_\rho^k)^{\frac{1}{2}}$ is used. With (D.3) and

$$\overline{U}_{1,k}^*(\overline{U}_{1,k}^*)^\top + \overline{U}_{2,k}^*(\overline{U}_{2,k}^*)^\top = I_{n_r}, \tag{E.2}$$

(E.1a) is rewritten as

$$\begin{aligned}(\boldsymbol{\Sigma}_\rho^k)^{-\frac{1}{2}}r_d^* = \Big[&\left(1+\nu\lambda_{i,k}^*\right)^{-1}\overline{U}_{1,k}^*(\overline{U}_{1,k}^*)^\top\\ &+\overline{U}_{2,k}^*(\overline{U}_{2,k}^*)^\top\Big](\boldsymbol{\Sigma}_\rho^k)^{-\frac{1}{2}}\mathbf{F}_\rho^k\theta.\end{aligned} \tag{E.3}$$

Then, from (D.2), (D.3) and (E.3), (E.1b) becomes

$$\begin{aligned}\|r_d^*\|^2_{\mathbf{W}_\rho^{k*}} &= \lambda_{i,k}^*(r_d^*)^\top(\boldsymbol{\Sigma}_\rho^k)^{-\frac{1}{2}}\overline{U}_{1,k}^*(\overline{U}_{1,k}^*)^\top(\boldsymbol{\Sigma}_\rho^k)^{-\frac{1}{2}}r_d^*\\ &= \lambda_{i,k}^*(1+\nu\lambda_{i,k}^*)^{-2}\left\|\mathbf{F}_\rho^k\theta\right\|^2_{(\boldsymbol{\Sigma}_\rho^k)^{-1}} = 1.\end{aligned} \tag{E.4}$$

Using (42), (D.2), (E.2), (E.3), (E.4), we have

$$\begin{aligned}&p^*(\tau_{k,\rho}(\theta),\gamma) = p(\theta,\mathbf{W}_\rho^{k*})\\ &= \left\|(\boldsymbol{\Sigma}_\rho^k)^{-\frac{1}{2}}r_d^* - (\boldsymbol{\Sigma}_\rho^k)^{-\frac{1}{2}}\mathbf{F}_\rho^k\theta\right\|^2\\ &= \left\|\left(\left(1+\nu\lambda_{i,k}^*\right)^{-1}-1\right)\overline{U}_{1,k}^*(\overline{U}_{1,k}^*)^\top(\boldsymbol{\Sigma}_\rho^k)^{-\frac{1}{2}}\mathbf{F}_\rho^k\theta\right\|^2\\ &= \left(\left(1+\nu\lambda_{i,k}^*\right)^{-1}-1\right)^2\left\|\mathbf{F}_\rho^k\theta\right\|^2_{(\boldsymbol{\Sigma}_\rho^k)^{-1}}\\ &= \left(\left\|\mathbf{F}_\rho^k\theta\right\|_{(\boldsymbol{\Sigma}_\rho^k)^{-1}} - \sqrt{\frac{\operatorname{rank}(\mathbf{F}_\rho^k)}{\gamma}}\right)^2\end{aligned}$$

which proves the second equality in (48).

F Proof of Corollary 5

Let $\mathbb{H}_0^G : \mathbf{r}_\rho^k \in \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_\rho^k)$ and $\mathbb{H}_1^G : \mathbf{r}_\rho^k \in \mathcal{N}(\mathbf{F}_\rho^k\theta, \boldsymbol{\Sigma}_\rho^k)$ denote the two hypotheses for the standard GLR test

(49). In the fault-free case, the GLR test statistic $T(\mathbf{r})$ in (49) follows the chi-squared distribution $\chi^2_{\text{rank}(\mathbf{F})}$, then we have

$$\mathbb{E}\left(T(\mathbf{r}_\rho^k)|\mathbb{H}_0^G\right) = \text{rank}(\mathbf{F}_\rho^k),\ \mathbb{P}\left(T(\mathbf{r}_\rho^k) > T_{th}(\gamma)|\mathbb{H}_0^G\right) = \gamma.$$

Then, the Markov inequality gives

$$\gamma = \mathbb{P}\left(T(\mathbf{r}_\rho^k) > T_{th}(\gamma)|\mathbb{H}_0^G\right) < \frac{\mathbb{E}\left(T(\mathbf{r}_\rho^k)|\mathbb{H}_0^G\right)}{T_{th}(\gamma)} = \frac{\text{rank}(\mathbf{F}_\rho^k)}{T_{th}(\gamma)},$$

which is rewritten as

$$T_{th}(\gamma) < \frac{\text{rank}(\mathbf{F})}{\gamma}. \tag{F.1}$$

In the faulty case, the GLR test statistic $T(\mathbf{r})$ in (49) follows the noncentral chi-squared distribution $\chi^2_{\text{rank}(\mathbf{F})}(\tau^2_{k,\rho}(\theta))$, with the noncentrality parameter $\tau^2_{k,\rho}(\theta)$ defined in (48a). With the GLR test threshold $T_{th}(\gamma)$, the corresponding fault detection rate is expressed as

$$\begin{aligned} P_D^{\text{GLR}}(\tau_{k,\rho}(\theta),\gamma) &= \mathbb{P}\left(T(\mathbf{r}_\rho^k) > T_{th}(\gamma)|\mathbb{H}_1^G\right) \\ &= 1 - Q_{\chi^2_{\text{rank}(\mathbf{F})}(\tau^2_{k,\rho}(\theta))}(T_{th}(\gamma)). \end{aligned} \tag{F.2}$$

According to (43) and (49), the distributionally robust test in (44) can be expressed as

$$J^*(\mathbf{r}_\rho^k) = \frac{\gamma T(\mathbf{r}_\rho^\mathbf{k})}{\text{rank}(\mathbf{F}_\rho^k)}. \tag{F.3}$$

Then, we derive the following inequality to prove the worst-case FDR is lower than the nominal FDR of the GLR test:

$$\begin{aligned} &P_D^*(\tau_{k,\rho}(\theta),\gamma) = \inf \mathbb{P}\left(J^*(\mathbf{r}_\rho^k) > 1|\mathbb{H}_1\right) \\ &= \inf \mathbb{P}\left(T(\mathbf{r}_\rho^k) > \frac{\text{rank}(\mathbf{F}_\rho^k)}{\gamma}\middle|\mathbb{H}_1\right) \\ &< \inf \mathbb{P}\left(T(\mathbf{r}_\rho^k) > T_{th}(\gamma)\middle|\mathbb{H}_1\right) < \mathbb{P}(T(\mathbf{r}_\rho^k) > T_{th}(\gamma)|\mathbb{H}_1^G) \\ &= P_D^{\text{GLR}}(\tau_{k,\rho}(\theta),\gamma), \end{aligned}$$

where the second equality relies on (F.3), and the first inequality in the third row comes from (F.1).

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